Communicative Efficiency in Adjective Ordering: Hierarchical Composition and Incremental Pragmatics in Reference Production

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Abstract

Adjective ordering preferences, such as placing size before colour, have been explained by two accounts of communicative efficiency. Hierarchical composition favours placing less subjective adjectives closer to the noun, where they restrict the interpretation of more subjective ones; incremental pragmatics favours placing more discriminating adjectives earlier, where they help the listener narrow down the referent sooner. However, the two accounts can favour opposing orders when the scene makes the less subjective adjective the more discriminating one. Which order speakers produce depends on what they evaluate: the complete description as a whole, or each adjective incrementally? The answer may lie in between. We develop Rational Speech Act models of adjective production that formalise both accounts and span this range, from a global to an incremental speaker. Between these endpoints, a plan-guided speaker chooses each adjective for its immediate usefulness while looking ahead to the complete description, so that both benefits can shape the order. We found that the plan-guided speaker best predicted German production data, whereas modelling hierarchical composition added no predictive benefit. Simulations nevertheless showed that composition can shape adjective order, but only under restricted conditions. Together, these findings motivate an account of boundedly rational communication in which partial planning supports good-enough production: speakers anticipate the communicative value of a description while resolving individual word choices incrementally.

Keywords: adjective ordering, production planning, reference production, Rational Speech Act, Bayesian cognitive modelling

1 Introduction

Speakers can begin an utterance before every part of it has been prepared and continue planning during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014). Adjective ordering shows the consequences of such incremental planning particularly clearly, although it has rarely been studied from this perspective. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Sproat and Shih, 1991; Scontras, 2023). Yet the first adjective is also the listener's first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Trotzke and Wittenberg, 2019; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

Do speakers decide on an order before they begin, or does the order emerge as they speak? Because adjectives are produced incrementally, both are possible. A speaker who has planned the complete description knows whether it will contain size, colour, or both, and can then decide how to order these properties. A speaker who plans less may instead begin with the property that is most salient or most discriminating in the scene, so that the order follows from which adjectives are added next. In the first case, order is a decision about a planned set of adjectives; in the second, it is a by-product of selection. Which order a speaker produces may thus depend on how much of the description is planned in advance.

Two accounts of communicative efficiency address the tension between conventional and context-dependent efficient order, one based on hierarchical composition and one on incremental interpretation. More subjective adjectives tend to occur farther from the noun, yielding a size-first preference for size and colour (Scontras, Degen, and Goodman, 2017; Trainin and Shetreet, 2021). The first account connects this pattern to hierarchical semantic composition: the noun-adjacent adjective restricts the context in which an outer adjective is interpreted

(Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Goodman, 2019; Scontras, Bar-Sever, et al., 2020). In *the big blue sticker*, *blue* combines with *sticker* before *big*, potentially narrowing the comparison class against which size is evaluated. Because colour is less subjective than size, evaluating *big* among the blue objects can reduce uncertainty about its interpretation.

During incremental production, an adjective's usefulness also depends on when it becomes available in the surface sequence. Listeners use partial linguistic input and visual context to restrict candidate referents before a noun phrase is complete (Tanenhaus et al., 1995; Sedivy et al., 1999). Placing the more discriminating property first can facilitate this process, favouring colour-first order when colour identifies the target sooner (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Hierarchical composition and surface production thus provide two directions in which adjective order can support efficient reference. The two accounts thus presuppose the two ways in which an order can arise: the compositional benefit of size-first order depends on the complete phrase, whereas the incremental benefit of colour-first order depends on each adjective as it is produced.

Both accounts compare two orders of the same adjectives, whereas in production the available orders depend on which properties are selected and how much information the description includes. In a colour-sufficient scene, *the blue sticker* may complete the description, so competition between size–colour and colour–size orders may never arise. In a size-sufficient scene, speakers may still produce *the big blue sticker*, including colour even when size alone identifies the target. Colour is included redundantly more often than relative adjectives such as size (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). Such redundancy can itself arise in the two ways described above: a speaker may begin with salient colour before establishing that size is needed (Pechmann, 1989), or include colour because it helps the listener find the target (Rubio-Fernández, 2016; Degen et al., 2020). Reference planning therefore involves selecting properties for mention as well as arranging them (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). Selection and ordering together make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome.

The Rational Speech Act (RSA) framework can predict this distribution by relating speaker choice to an utterance's usefulness to a listener and its production cost (Frank and Goodman, 2012; Goodman and Frank, 2016; Degen, 2023). Schlotterbeck and Wang (2023) combined the two efficiency accounts in an incremental RSA model to capture size-first preferences and their modulation by referential context. That account compared two fixed adjective orders, leaving unresolved how much speakers plan and how planning shapes the content and length of a description. We extend this account by comparing fully planned (global) production, fully incremental production (Cohn-Gordon, Goodman, and Potts, 2019), and an intermediate hypothesis that we call *plan-guided incremental production*. Under this hypothesis, a partial plan of the complete description guides each adjective, while immediate cues still compete. The global and fully incremental endpoints correspond to the two ways in which an order can arise. A free parameter determines the degree of planning between these endpoints.

Our central research question is: *How much of a description do speakers plan when selecting and ordering adjectives?* We combine two German slider-rating studies and a production study with Bayesian predictive model comparison and controlled simulation. The slider studies measure order preferences when both complete descriptions are given; the production study shows which content, length, and order speakers produce when they must plan the description themselves. We ask which degree of planning best predicts these descriptions and whether narrowing the comparison class for size, as proposed by the compositional account, improves prediction. The simulation then examines how the communicative advantage of an order changes when shorter descriptions are also available.

Figure 1 summarises the studies, the models compared, and the main findings. Section 2 introduces the task and response space, followed by the models and their degrees of planning in Section 3. Section 4 presents the empirical evidence and predictive comparisons; Section 5 examines the conditions for a communicative ordering advantage. Section 6 discusses the implications for production planning, reference production, and semantic composition.

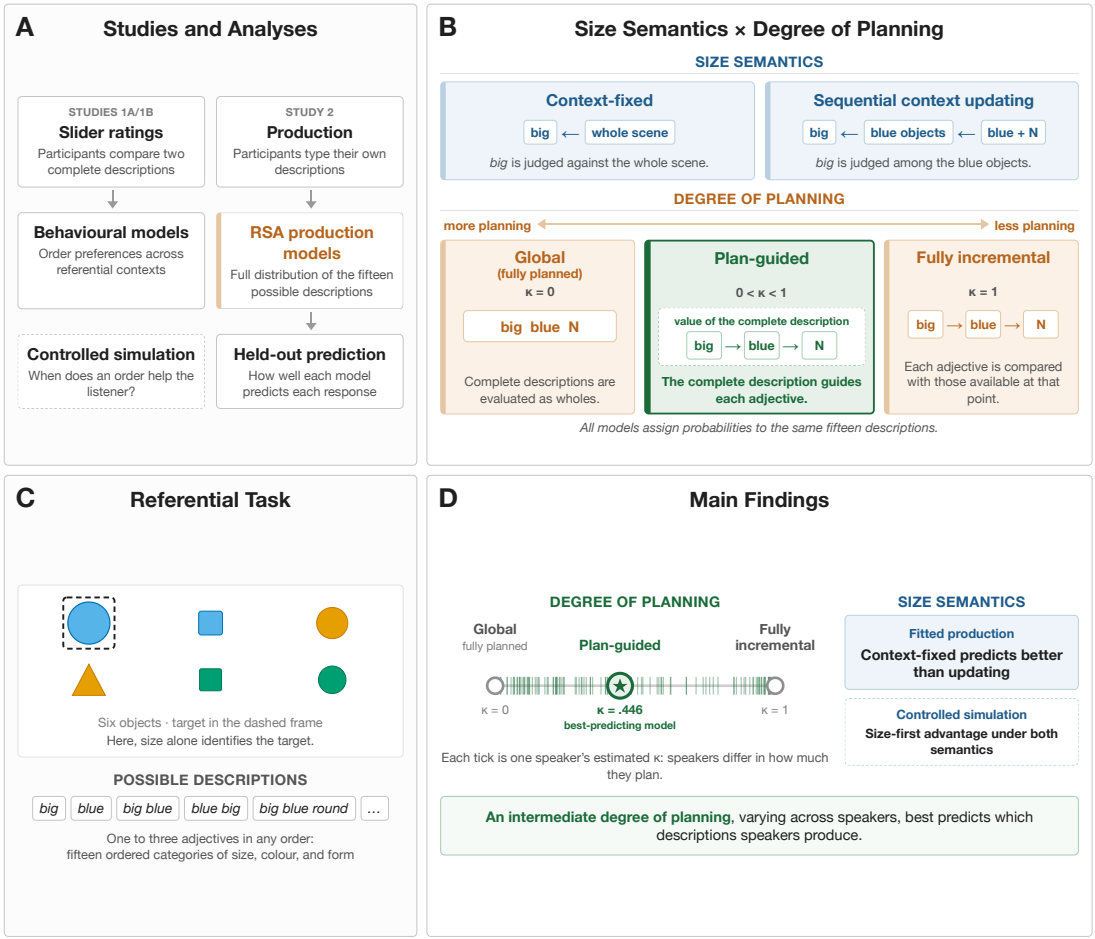


Figure 1: Overview of the studies, models, and findings. (A) The slider-rating studies are analysed with behavioural models; the production study is used to compare RSA production models by held-out prediction, and controlled simulations examine when an order helps the listener. (B) The production models cross two size semantics with three degrees of planning. The comparison class for *big* is either the whole scene or the objects matching the noun-adjacent adjective; production ranges from fully planned (global, $\kappa = 0$) through plan-guided ($0 < \kappa < 1$) to fully incremental ($\kappa = 1$). (C) An example display in which size alone identifies the target, with examples of the fifteen possible descriptions. (D) Plan-guided production predicts the production data best (estimated $\kappa = .446$); ticks show each participant's estimated κ. Context-fixed semantics predicts better than updating, while simulations show a size-first advantage under both.

2 Referential Task

All three studies used the same referential task. The task was presented as a reference game about collectible stickers: a partner saw the same stickers in a different order and lacked the target frame, making spatial descriptions uninformative. On each critical trial, participants saw six objects with one framed target. The noun was always *Aufkleber* ('sticker'), and objects varied in size, colour, and form. The target was always the largest object of its colour and form, so it could always be described as *big*.

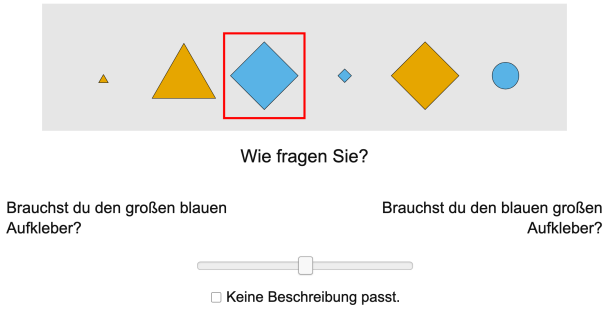
In Study 1a and its direct replication, Study 1b, participants used a continuous slider to indicate which of two phrases they preferred, such as *den großen blauen Aufkleber* and *den blauen großen Aufkleber* ('the big blue sticker' and 'the blue big sticker'). In Study 2, participants saw the sentence frame *den* ___ *Aufkleber* and typed adjectives into it to describe the target to their partner. This response format allowed us to observe which properties they mentioned, how many adjectives they used, and how they ordered them. Participant samples, behavioural results, and the fitting procedure are reported in Section 4; full experimental details appear in Section A.

Each study used a 3 (adjective pair: size–colour, size–form, colour–form; within-subjects) × 3 (referential context: first property sufficient, second property sufficient, both necessary; within-subjects) × 2 (size discriminability: high, low; between-subjects) mixed design. Referential context changed which properties were needed to identify the target; in size–colour trials, for example, size alone (size sufficient), colour alone (colour sufficient),

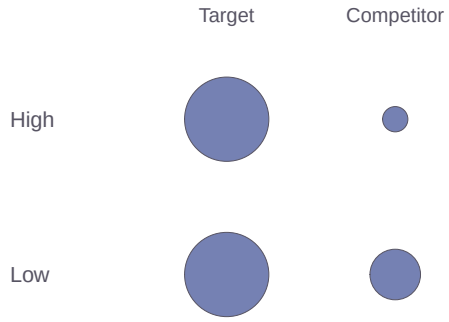
or both properties (both necessary) identified the target. Target–competitor size differences were more pronounced in the high-discriminability group than in the low-discriminability group (Figure 2). Referential context changes whether size is needed; discriminability changes how easily the size difference can be seen, while the adjective *big* stays the same. The behavioural analyses focus on size–colour trials, the classic contrast in the ordering literature and our motivating example. The production models are fitted to all three adjective pairs, with colour–form trials providing a baseline without a subjective adjective.

A production response could contain a single adjective, such as *blue*; two adjectives, such as *big blue*; or three, such as *big blue round*. Coding each adjective as size (S), colour (C), or form (F) gives fifteen ordered categories: three single adjectives, six two-adjective sequences, and six three-adjective sequences. All production models assign probabilities to these fifteen categories. They are fitted to Study 2; the slider studies provide separate evidence about explicit preferences between two competing orders.

A



B



C

Design dimension	Levels
Adjective pair	Size–colour; size–form; colour–form (within-subjects)
Referential context	First property sufficient; second property sufficient; both necessary (within-subjects)
Size discriminability	High; low (between-subjects)

Figure 2: Experimental design and materials. Panel A reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders. Panel B illustrates the size-discriminability manipulation schematically using recorded stimulus levels: the target–competitor size gap is larger under high discriminability and smaller under low discriminability. Panel C summarises the $3 \times 3 \times 2$ factorial design.

3 A Plan-Guided Production Model

Consider a scene in which colour alone identifies the target. A fully planned speaker compares complete descriptions, such as *blue*, *big blue*, and *blue big*, so a first adjective is likely when it begins good descriptions. A fully incremental speaker evaluates the first adjective on its own, so *blue* is likely because it immediately singles out the target, whatever follows. Both speakers pursue the same goal of helping the listener identify the target; they differ only in how far ahead they look. Which of them do human speakers resemble, or do they lie in between?

The Rational Speech Act (RSA) framework models a speaker who selects utterances by reasoning about how a literal listener would interpret them, favouring utterances that lead the listener to the intended referent at low cost (Frank and Goodman, 2012; Goodman and Frank, 2016). Standard RSA speakers evaluate complete utterances, whereas incremental variants evaluate one word at a time (Cohn-Gordon, Goodman, and Potts, 2019). Our model keeps this listener-oriented reasoning but treats the choice between these two forms of evaluation as a matter of degree, estimated from the data. Because planning cannot be observed directly in typed descriptions, the model infers it from the distribution of descriptions that speakers produce: which properties they mention, how many adjectives they use, and how they order them. The models also differ in whether the comparison class for size is updated as adjective meanings compose from the noun outwards. Panel B of Figure 1 shows these two dimensions.

3.1 Literal Listener and Meaning Function

All models use the observed six-object display. Size has a graded meaning: an object counts as *big* to the degree that it is larger than the other objects in its comparison class (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019). Colour meanings are graded as well: objects of the named colour receive most, but not all, of the weight. Form is treated as uninformative in the literal semantics; speakers' use of form is captured by salience and the length and selection terms. These semantic settings are shared across the production architectures.

For every complete or partial adjective string, the literal listener L interprets the adjectives by hierarchical composition, applying modification from the noun outwards. For example, in *big blue sticker*, *blue* modifies *sticker* before *big* is interpreted. Let $c_1, \ldots, c_T$ denote the modifiers of a string in this order, so that c_1 is the adjective next to the noun. Starting from a uniform prior over the objects, $C_0 = P_0$, composition updates a distribution over objects at each step:

$$C_j(o) \propto C_{j-1}(o)\, [\![c_j]\!]_{\rho_j}(o), \qquad j = 1, \ldots, T, \tag{1}$$

where $[\![c_j]\!]_{\rho_j}$ is the modifier's meaning under the selected semantic regime. The final state gives the literal listener's distribution over objects.

Hierarchical composition affects order only if an inner adjective can change how an outer adjective is interpreted. The two semantic regimes differ in exactly this respect. Under *context-fixed* semantics, the comparison class for size remains the whole scene. Under *sequential context updating*, the comparison class is recomputed after each modifier, so that in *big blue*, *big* is judged among objects weighted by how well they match *blue*. Because colour meanings are graded, this restriction is partial. When it makes the target more distinctive in size, updating can make size-first order more helpful to the listener. The threshold construction and semantic parameters appear in Sections B and G.

Interpretation proceeds from the noun outwards, but production proceeds from left to right. The speaker evaluates the information available after each adjective in the surface sequence. Let $u_{\leq t} = w_1 \ldots w_t$ be the prefix comprising the first t adjectives, and $P_0(o \mid X) = 1/6$ the object prior in display X. Each prefix is interpreted from this prior, applying each of its modifiers once in noun-outward order. For *big blue*, the first evaluation uses *big* alone, and the second uses the composed meaning of *big blue*. Under context-fixed semantics, the listener after a prefix is

$$L(o \mid u_{\leq t}, X) \propto P_0(o \mid X) \prod_{j=1}^{t} [\![w_j]\!]_{\text{fix}}(o). \tag{2}$$

Under updating, the comparison class for *big* therefore changes as the prefix grows from *big* to *big blue*. The probability that this listener assigns to the target is the communicative value of the prefix.

3.2 Incremental Speaker and Plan-Guided Production

The three production architectures share a *prefix evaluation* rule: the speaker evaluates how the adjectives in an utterance guide the listener to the target at each adjective position. Accumulating these evaluations means that a description gains value when it identifies the target early, not only when it is complete. The production architecture determines how this accumulated utility is compared with alternatives: across complete descriptions, or among the adjectives available at each point. At position t, the utility of candidate adjective a for target o^* is

$$U_t(a \mid u_{<t}, o^*, X) = \alpha \log L(o^* \mid u_{<t} + a, X) + \lambda_{\text{salience}} z_a, \tag{3}$$

where α controls sensitivity to listener success and z_a represents the visual salience of the candidate property. The two terms capture what can make an adjective attractive as soon as it is produced: it discriminates the target, or it is visually salient. For a completed utterance u, the model combines the accumulated utility of its selected adjectives with a term for competition among alternatives at each adjective position:

$$U^{\text{complete}}(u, o^*, X) = \sum_t U_t(w_t \mid u_{<t}, o^*, X), \tag{4}$$

$$\mathcal{N}^{\text{local}}(u, o^*, X) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}, o^*, X), \tag{5}$$

where $\mathcal{A}(u_{<t})$ contains the adjective types still available after the current prefix. The complete-utterance utility U^{complete} is the sum of these prefix evaluations and is shared by all three production architectures. The local-normalisation term $\mathcal{N}^{\text{local}}$ records the utility of the competing adjectives at each position. When adjectives

compete at a given point, a selected adjective receives lower probability if other available adjectives are also useful there. For example, *big blue* and *blue big* contain the same properties but face different competitors at the first position. When colour alone identifies the target, *blue* is a strong competitor, so *big* is less likely to come first. Two completed utterances with similar U^{complete} can therefore receive different probabilities because they pass through prefixes with different degrees of competition. These two terms correspond to the two influences in plan-guided production: the value of the complete description and the competition among adjectives at each point.

The probability of a completed utterance is

$$S_\kappa(u \mid o^*, X) \propto \exp\left\{U^{\text{complete}}(u, o^*, X) - \kappa \mathcal{N}^{\text{local}}(u, o^*, X) + \beta\Omega(u) + T(u, X)\right\}. \tag{6}$$

The parameter κ determines how strongly competition among the adjectives available at each point contributes to the probability of a completed utterance. With $\kappa = 0$, this competition drops out, and the *global* speaker compares only the fifteen complete descriptions. With $\kappa = 1$, the *fully incremental* speaker produces each adjective by comparing it with the adjectives available at that point, without looking ahead; the probability of a description then combines these local choices with the shared terms below.[1] Intermediate values define the *plan-guided* speaker, in which both comparisons contribute. Lower values of κ therefore correspond to a greater degree of planning. Even the global speaker still evaluates each prefix; it differs only in comparing complete descriptions.

Besides informativeness and salience, the descriptions are shaped by tendencies that are not specific to planning: speakers follow conventional orders, avoid unnecessarily long descriptions, prefer a single adjective when it suffices, and add properties when size is hard to see. Our account does not aim to derive these tendencies, but a model that ignored them would misattribute their effects to the degree of planning. Following common practice in Bayesian cognitive modelling, we include them as auxiliary components that link the model to the task (Lee and Wagenmakers, 2013; Franke, 2016): a fixed order score $\Omega(u)$ from a German language model, weighted by β, and length and selection terms $T(u, X)$. Their parameters are estimated jointly with κ, so uncertainty about them is carried into the inference about planning. Because these components are identical across architectures, the architectures differ only in how descriptions are evaluated. Their specification and development history appear in Sections B and D.

3.3 Matched Model Comparisons

Crossing the three production architectures with the two semantic regimes gives six matched models (Table 1), which share all other components. All also allow participant variation in pragmatic optimality α_i and stable-order weight β_i. The comparison asks whether an intermediate degree of planning improves prediction over both endpoints, whether updating the size comparison class adds further predictive value, and whether speakers differ in how much they plan.

Table 1: Production models compared. Each model is fitted with context-fixed and with context-updating semantics. The first three rows form the six matched models; the last row adds the two refinements with participant-specific degrees of planning.

Production architecture	Degree of planning
Global (fully planned)	$\kappa = 0$
Plan-guided	Shared κ, estimated between 0 and 1
Fully incremental	$\kappa = 1$
Plan-guided with participant variation	κ_i, estimated for each participant

4 Empirical Evidence and Model Comparison

This section evaluates the production models against the German data. The slider studies show how referential context shapes preferences between two complete orders; the production study additionally shows which properties speakers select and how long their descriptions are, and it therefore provides the data for model comparison. After summarising the samples and observed patterns, we ask which degree of planning best predicts production, whether planning varies across speakers, and whether updating the comparison class for size improves prediction, before checking the predictions of the best-predicting model. Full experimental methods, behavioural analyses, and secondary results appear in Section A.

[1] This contrast corresponds to that between locally and globally normalised sequence models in natural language processing (Lafferty, McCallum, and Pereira, 2001; Andor et al., 2016).

4.1 Studies and Observed Responses

All studies were conducted online with native German speakers recruited via Prolific. Study 1a used the dataset reported by Schlotterbeck and Wang (2023); Study 1b replicated the task and materials in an independent sample. Study 2 elicited production, with participants from either slider study ineligible. Table 2 distinguishes the size–colour subsets used for the behavioural analyses from the complete production dataset used to compare the RSA models. The first author coded production responses into the fifteen categories in Section 2. Full exclusion and coding criteria are reported in Section A. In particular, the exclusion of production participants with more than 80% identical responses selects a sample that varies its descriptions across trials.

Table 2: Retained samples and observations used in each analysis. The full production comparison includes size–colour, size–form, and colour–form trials.

Study	Participants	Responses	Analysis
1a: Slider ratings	134	3,485	Size–colour behavioural contrasts
1b: Slider replication	104	2,786	Size–colour behavioural contrasts
2: Production	113	3,034	Size–colour behavioural contrasts
		9,100	Full production-model comparison

Explicit ordering preferences. Both slider studies showed the same relation between referential context and adjective-order preference: size-first ratings were strongest when size alone identified the target and weakest when colour alone did (Figure 7). The joint posterior probability of the ordering size-sufficient $>$ both-necessary $>$ colour-sufficient exceeded .999 in Study 1a and was .999 in Study 1b. A small size-first preference remained in colour-sufficient displays: the estimated centred rating was 0.054 (95% credible interval [CrI] $[0.003, 0.109]$) in Study 1a and 0.057 (95% CrI $[0.001, 0.116]$) in Study 1b, on a scale from -0.5 to 0.5 with zero indicating neutrality. The interaction between size discriminability and context observed in Study 1a was not clearly replicated in Study 1b. The replicated result therefore concerns the context gradient and the residual size-first preference; the evidence for how discriminability modifies that gradient was mixed.

Adjective selection, length, and order. The production distribution in Figure 3 shows how referential context shaped what speakers chose to mention. Both-necessary scenes generally elicited longer descriptions, whereas one-property-sufficient scenes more often elicited a single adjective. Colour-sufficient scenes were commonly resolved with colour alone, leaving no size–colour ordering decision to observe on those trials. Conditional on producing both size and colour, size-first descriptions remained more frequent and colour-first descriptions were comparatively rare.

Perceptual reliability also affected the descriptions. Within size-sufficient displays, lower discriminability increased the estimated probability of redundant adjective use by 23.4 percentage points (95% CrI $[15.1, 31.8]$). Within both-necessary displays, higher discriminability increased the probability of a size-initial response by 14.1 percentage points (95% CrI $[3.6, 24.2]$). Both effects varied credibly across contexts in the planned interaction contrasts (Table 7). Redundancy means mentioning more properties than required for unique reference; size-initiality includes responses consisting of size alone and therefore reflects selection as well as order. The production models must account for this joint distribution of content, length, and order, including the redundant descriptions whose frequency varies across participants.

4.2 Inference and Predictive Comparison

We fitted the production models to the same 9,100 annotated responses from 113 participants in Study 2, including size–colour, size–form, and colour–form trials. Each response was represented as one of the fifteen ordered adjective categories used by every model. Bayesian inference was performed in NumPyro (Phan, Pradhan, and Jankowiak, 2019). The slider ratings were analysed separately with behavioural hierarchical models, as specified in Section A. The six matched production models and two participant-specific refinements follow the restrictions introduced in Section 3.

Approximate leave-one-out expected log predictive density (ELPD) measures how much probability each model assigns to held-out responses (Vehtari, Gelman, and Gabry, 2017). Higher ELPD indicates better prediction of the complete utterance distribution. The predictive target is another response from a participant represented in the dataset, with that participant's remaining trials available to estimate their participant-specific parameters.

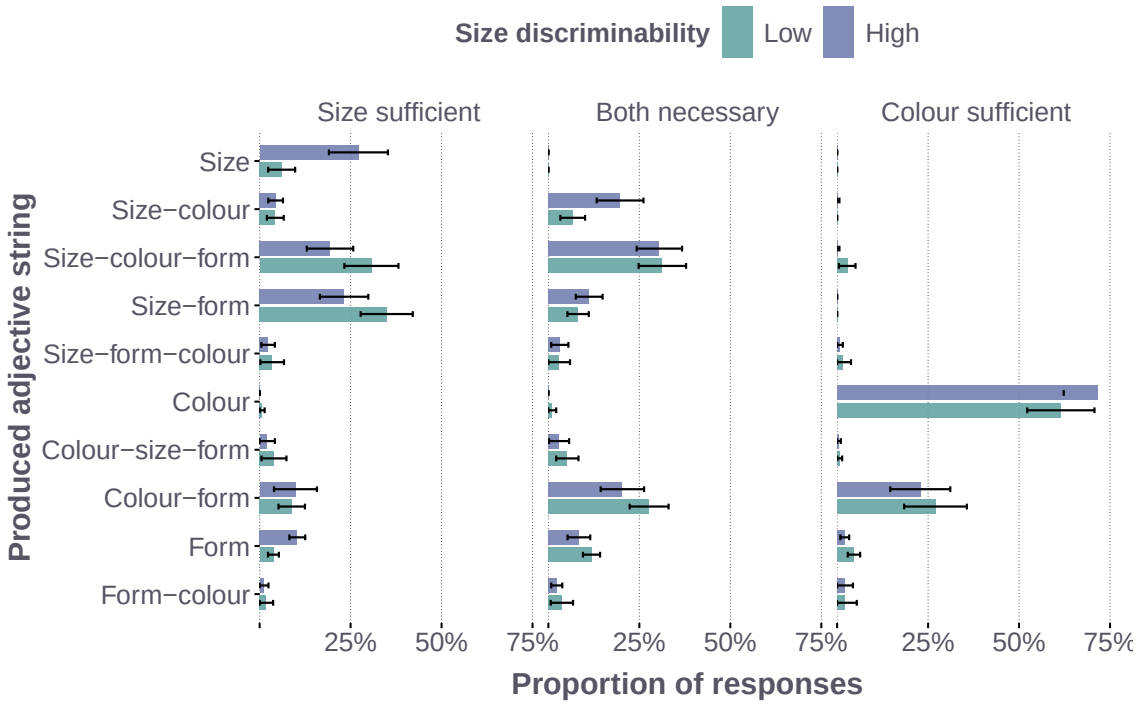


Figure 3: Ordered response distribution across the 3,034 size–colour production trials. Bars show mean participant-level proportions by referential context and size discriminability, with 95% confidence intervals. Adjective labels preserve surface order; the figure shows responses reaching at least 2% in one condition.

Model development used these data, and response-wise leave-one-out prediction evaluates the specifications selected through that development. Uncertainty in paired model comparisons is estimated with a participant-level Bayesian bootstrap and reported as 95% credible intervals. The same predictive target and uncertainty calculation are used throughout the architecture and hierarchy comparisons. The inference procedure and numerical diagnostics appear in Section C.

4.3 Production Architecture

The Bayesian model comparison favoured plan-guided production over both global and fully incremental production (Figure 4, Panel A). Averaged across semantic regimes, the plan-guided model had higher predictive accuracy than the global model ($\Delta\text{ELPD} = 256.9$, 95% CrI $[71.8, 404.7]$) and the fully incremental model ($\Delta\text{ELPD} = 548.5$, 95% CrI $[349.8, 795.1]$). These contrasts supported the plan-guided production hypothesis that complete-utterance evaluation and competition during incremental adjective choice jointly shape the response distribution. In the context-fixed model, the posterior mean shared incremental-choice weight was $\kappa = .446$, with a 95% CrI of $[.407, .486]$ (Panel B). The estimate lay between the global and fully incremental endpoints, indicating an intermediate contribution of local competition within the fitted architecture.

The predictive differences were most apparent in form-involving trials, where colour and form competed for the initial position. Colour-initial responses accounted for 81.7% of the plan-guided model's ELPD advantage over the average of the global and fully incremental models. Different opening adjectives leave different continuations available, allowing competition at later positions to affect the probabilities of the completed descriptions. The advantage therefore extended beyond the motivating size–colour contrast to the selection and ordering of properties in the broader response space.

4.4 Plan-Guided Production and Participant Variation

Allowing incremental-choice weights to vary across participants yielded a credible improvement over the otherwise matched model with a shared weight. Under context-fixed semantics, the participant-specific model gained 364.7 ELPD (95% CrI $[248.2, 521.0]$); the gain under sequential context updating was similar. The posterior mean population location remained intermediate between the global and fully incremental endpoints ($\mu_\kappa = .472$, 95%

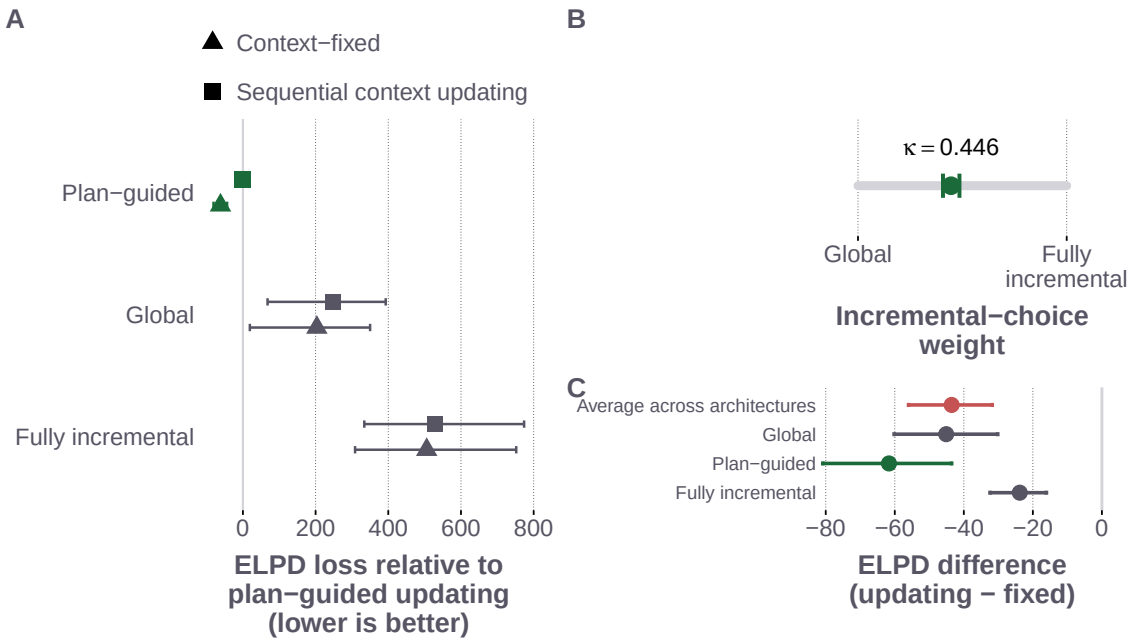


Figure 4: Matched comparisons of production architecture and sequential context updating. Panel A reports predictive loss for all six matched models relative to the plan-guided model with sequential context updating; lower values indicate better leave-one-out prediction. Panel B places the shared incremental-choice weight κ between the global and fully incremental endpoints. Panel C shows updating-minus-fixed ELPD for each production architecture and their average. Intervals in Panels A and C are 95% credible intervals from the participant-level Bayesian bootstrap; Panel B shows the 95% posterior credible interval for κ. The vertical line in Panel C marks zero.

CrI $[.337, .618]$), alongside substantial variation in participant-specific weights. Thus, the improvement came from accommodating differences across participants while retaining an intermediate population location.

The predictive benefit was concentrated in two- and three-adjective utterances, whereas predictions of single-adjective responses became less accurate. The gains for longer utterances outweighed this loss and extended to three-adjective descriptions, which contain redundant information in the present task. Participant variation therefore improved the model's account of descriptions involving several adjective choices, including the selection and ordering of additional properties. Section E reports the response-length contrasts and participant-level diagnostics.

4.5 The Contribution of Semantic Updating

We next asked whether updating the size comparison class improves prediction of production. The comparison favoured context-fixed semantics over sequential context updating under all three production architectures (Figure 4, Panel C). The average updating-minus-fixed contrast was negative ($\Delta\text{ELPD} = -43.5$, 95% CrI $[-56.0, -31.8]$), indicating a credible predictive advantage for the context-fixed models. This advantage remained in the participant-specific plan-guided comparison.

Decomposing the participant-specific model comparison showed that the semantic difference chiefly concerned how much speakers said and which properties they mentioned. Context-fixed semantics improved predictions of utterance length ($\Delta\text{ELPD} = 21.5$, 95% CrI $[12.4, 30.9]$) and adjective content conditional on length ($\Delta\text{ELPD} = 31.1$, 95% CrI $[23.5, 39.2]$). For order conditional on adjective content, the estimated advantage was small and its interval included zero ($\Delta\text{ELPD} = 1.5$, 95% CrI $[-1.8, 4.9]$). The comparison therefore did not establish a clear predictive advantage for either semantic regime in conditional order.

Both models also showed response-specific discrepancies, including overprediction of three-adjective descriptions in size-sufficient size–form trials. Full contrasts, the score decomposition, and the residual analysis appear in Section C.3.

4.6 Predictive Adequacy of the Best-Predicting Model

Posterior predictive checks assessed how closely the model reproduced the observed context sensitivity of adjective selection, utterance length, and initial position. The plan-guided model with context-fixed semantics and participant-specific incremental-choice weights captured the broad context-dependent shifts in size-initial responses and redundant adjective use (Figure 5, Panel A). The comparison across all fifteen utterances and eighteen experimental conditions showed correspondence between observed and predicted proportions, alongside discrepancies for particular responses (Panel B).

The supplementary condition-wise checks made these patterns more explicit (Figure 13). In size–colour trials, the model reproduced frequent colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. It nevertheless overpredicted colour-only responses in colour-sufficient conditions and misestimated the balance between some two- and three-adjective descriptions. In size–form trials with both properties necessary and high size discriminability, it underpredicted size-initial responses (predicted: 45.2%; observed: 59.5%) and overpredicted form-initial responses (predicted: 13.5%; observed: 4.2%). The model therefore captured broad shifts in descriptive content while leaving errors in particular responses and initial-adjective choices. Descriptive measures of fit appear in Section C.

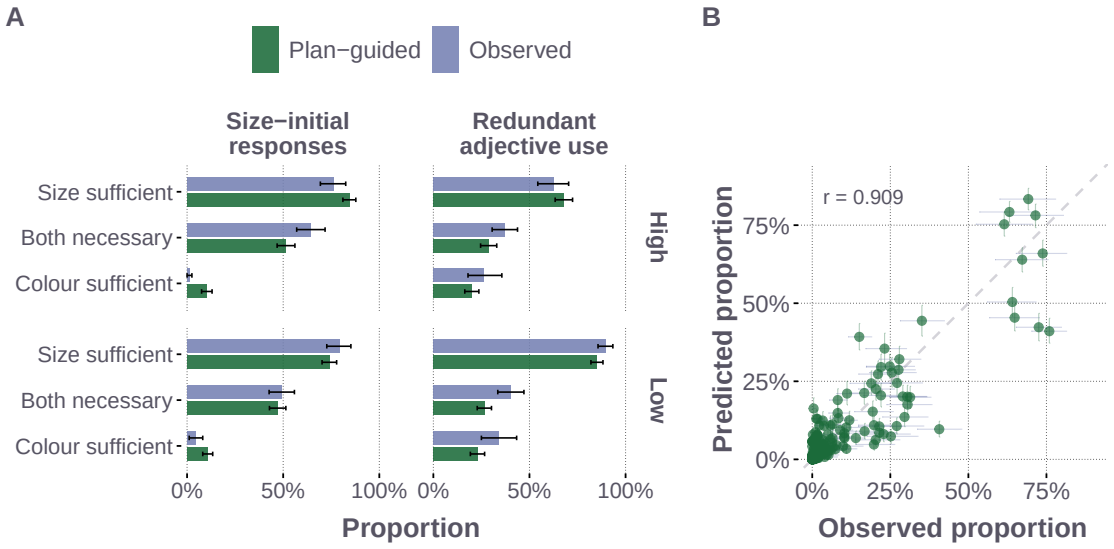


Figure 5: Posterior-predictive adequacy of the best-predicting plan-guided model. Panel A compares observed and posterior-predicted size-initial responses and redundant adjective use in size–colour trials, with referential context on the vertical axis and size discriminability in facet rows. Intervals show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions. Panel B compares all 270 response cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality.

These predictive comparisons leave open how a communicative advantage for size-first order arises under each semantic regime. The following simulation examines how this advantage arises and how it changes when adjective selection and utterance length enter the speaker's alternatives.

5 Simulation: Conditions for a Size-First Advantage

Earlier simulations found an average referential advantage for orders that place more subjective adjectives farther from the noun, motivating a pressure that could gradually stabilise ordering preferences across repeated language use (Franke, Scontras, and Simonic, 2019). Sequential context updating provides a mechanism for this advantage: the noun-adjacent adjective restricts the context in which an outer, more subjective adjective is interpreted. The production findings place this proposal within a broader set of choices, because speakers also decide which adjectives to include and how much to say. We therefore ask how the ordering advantage changes when the speaker evaluates information at each adjective position and can choose descriptions of different lengths.

5.1 Controlled Changes in Evaluation and Available Descriptions

The simulation reuses the same 24,000 randomly generated displays across all comparisons. Displays vary in object count and size distribution; the largest object is the target and is assigned blue. Varying the number of blue distractors changes how strongly colour restricts the comparison class for size. The scene-generation procedure and semantic parameter grid appear in Section H.

We introduce the production model's assumptions in controlled steps (Figure 6, Panel A). The first comparison gives a global speaker two alternatives, *big blue* and *blue big*, evaluated through the literal listener's target probability after the completed utterance. The second retains global choice ($\kappa = 0$) but introduces the prefix evaluation in Section 3.2, summing listener log probabilities at each adjective position. For *big blue*, the utility then includes the information supplied by *big* as well as by the completed description. Holding global choice fixed isolates the contribution of this prefix information. We next add the single-adjective alternatives *big* and *blue*, and finally expand to the fifteen utterances used in production. These four comparisons average over the same grid of 125 semantic parameter settings under each semantic regime.

A further comparison holds the fifteen utterances and production semantic constants fixed while varying the contribution of local competition: global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production (Panel B). All three architectures use prefix evaluation. Throughout these comparisons, pragmatic optimality is $\alpha = 1$, order, salience, length, and selection terms are zero, and final Laplace smoothing is omitted. Section H.1 gives the full specification.

The pragmatic listener L^{prag} applies Bayes' rule to the speaker distribution, using an equal prior over objects. An object receives more probability when a speaker intending that object would be more likely to produce the observed utterance. For each context, the size-first advantage compares the listener's target probability after the two orders:

$$\Delta_{\text{SF}} = L^{\text{prag}}(o^* \mid \text{“big blue”}) - L^{\text{prag}}(o^* \mid \text{“blue big”}). \tag{7}$$

Positive values favour size-first order. Other available descriptions can affect this measure because the listener reasons about the speaker's choice among them.

5.2 Conditions under which an Order Asymmetry Emerges

With global choice and complete-utterance evaluation, sequential context updating generated a mean size-first advantage of approximately 1.1 percentage points, while context-fixed semantics yielded no difference between the orders (Figure 6, Panel A). With fixed meanings, the two adjective orders have the same completed interpretation and therefore the same target probability under this evaluation rule. Updating breaks this symmetry because colour restricts the comparison class for size. This comparison recovers the proposed communicative contribution of hierarchical restriction.

Including information available at each adjective position changed the result. With only the two orders available, the global production model yielded a mean colour-first advantage under both semantic regimes: approximately 2.7 percentage points with fixed meanings and 1.8 with updating. The first adjectives can differ in informativeness even when the completed meanings coincide, allowing an ordering asymmetry under fixed semantics. Its direction, however, depends on the alternatives available for reference. Adding the single-adjective responses changed the mean advantage to size-first under both regimes, to approximately 2.3 and 2.8 percentage points, respectively. The speaker can now identify a referent with one adjective, changing how probability is distributed over the two-adjective descriptions and how the pragmatic listener interprets their use. Adjective selection and utterance length thus affect the communicative value of an order even when the two orders themselves remain unchanged.

Expanding to fifteen utterances retained the mean size-first advantage, at approximately 2.6 percentage points with fixed semantics and 3.1 with updating. At the semantic constants used in production, all three production architectures generated a mean size-first advantage under both semantic regimes (Figure 6, Panel B). The advantage was approximately 1.9 percentage points for global production, 1.5 for plan-guided production, and 1.0 for fully incremental production, with nearly identical values for fixed and updating semantics within each architecture. Increasing the contribution of local adjective competition reduced the magnitude of the advantage while preserving its direction in this comparison. Updating's additional contribution therefore depends on the semantic settings, and the size-first advantage in the broader utterance space extends across the three production architectures.

The simulation shows how the interpretation and selection of adjectives jointly determine the communicative advantage of an order. Semantic updating can generate an advantage when comparing completed meanings; evaluating information available at each adjective position and allowing shorter descriptions changes both the source and direction of that advantage. The effect of adding single-adjective alternatives gives a computational

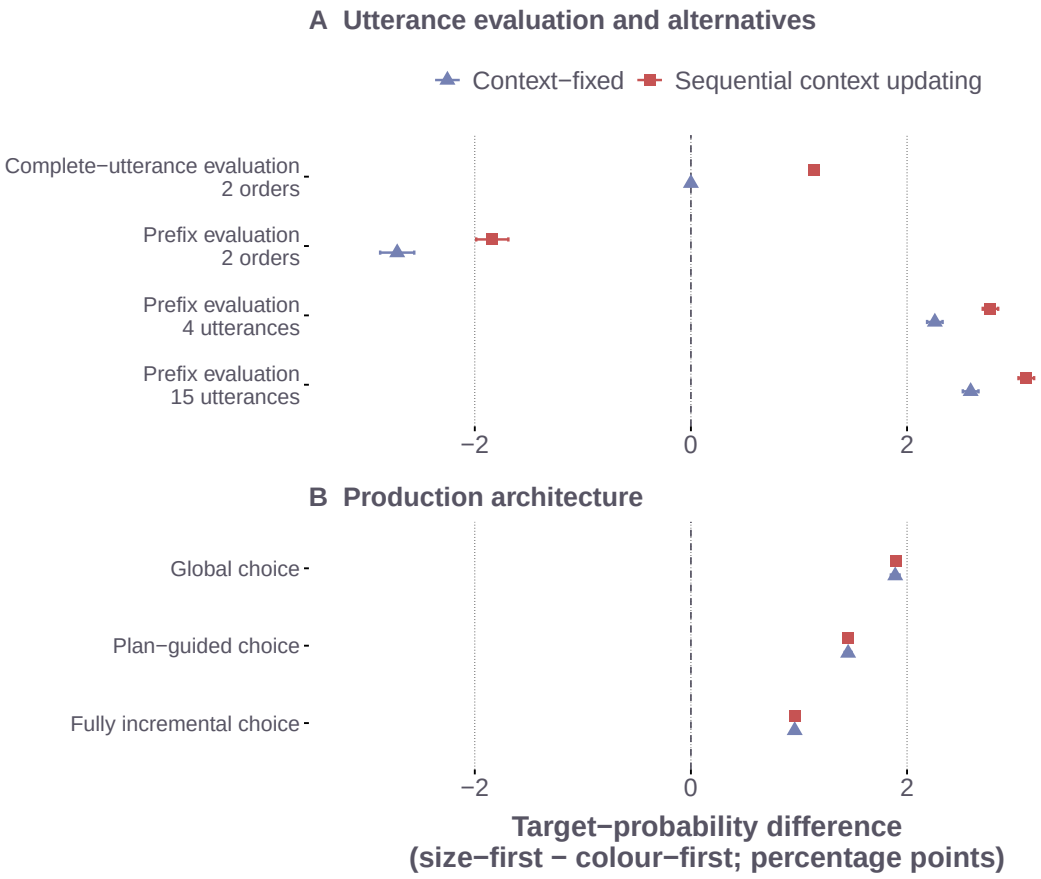


Figure 6: Size-first pragmatic target-recovery advantage across utterance evaluation, alternatives, and production architecture on the same 24,000 random scenes. Panel A holds global choice fixed and averages over the original semantic parameter grid. Complete-utterance evaluation uses only the final literal-listener target probability; prefix evaluation sums listener log probabilities at each adjective position. The four-utterance space adds single adjectives to the two orders; the fifteen-utterance space contains all production responses. Panel B holds the fifteen utterances and production semantic constants fixed while comparing global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production. All three use prefix evaluation. Points show scene-averaged target-probability differences; positive values favour size-first and negative values favour colour-first. Intervals are ± 1.96 Monte Carlo standard errors, calculated after averaging parameter settings within scenes.

basis for considering order within the broader production distribution: the option of using a shorter description affects how the listener interprets the same two-adjective utterances.

6 General Discussion

The present findings connect adjective ordering to the cognitive organisation of reference production. Stable size-first preferences coexist with context-sensitive choices of descriptive content and length. Among the matched production models, the best predictions combine evaluation of complete descriptions with continued influence from local adjective competition. Participant-specific weights further improve prediction, particularly for descriptions containing several adjectives. These findings support a plan-guided account of production in which speakers anticipate the communicative value of a description while resolving individual choices incrementally.

6.1 Bounded Planning and Redundant Descriptions

A useful description must be selected from alternatives that differ in content, length, and order. As the number of available properties increases, their combinations and orderings grow combinatorially, making exhaustive comparison increasingly demanding. The cognitive problem is how speakers preserve sensitivity to the communicative consequences of a description while limiting the alternatives and continuations they evaluate. Computational tractability constrains the processes and representations that can support this behaviour (Rooij, 2008); approximation alone provides no general guarantee of tractability (Kwisthout, Wareham, and Van Rooij, 2011).

Plan-guided production motivates a hypothesis about how planning could be organised under these constraints. An utterance-level plan could specify properties relevant to identifying the target, constraining the alternatives considered during incremental word choice. Local choices could remain sensitive to the developing description and the properties still available for mention. This organisation accords with evidence that conceptual planning and linguistic formulation can proceed with different scopes (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). It also connects to good-enough processing, in which representations can remain partial while serving the current task (Ferreira, K. G. D. Bailey, and Ferraro, 2002).

Redundant descriptions provide a case in which planning and communicative goals interact. A property may be selected because it becomes accessible early, because it helps the listener identify the target, or because both pressures favour its inclusion (Pechmann, 1989; Engelhardt, K. G. Bailey, and Ferreira, 2006; Rubio-Fernández, 2016; Degen et al., 2020). Readily available information can guide visual search or hedge uncertainty about another property (Rubio-Fernandez, 2019; Tourtouri, Delogu, and Crocker, 2021). The improvement from participant-specific incremental-choice weights suggests that the balance represented by the model varies across speakers. Its gains for longer descriptions, alongside poorer prediction of single-adjective responses, locate this improvement in choices involving several adjectives.

The present model specifies a distribution over completed utterances and evaluates the full response space. A process account must additionally explain how speakers generate candidate descriptions and allocate computation to evaluating them. A resource-rational approach would relate the benefit of further planning to its time and processing costs (Lieder and Griffiths, 2020). Manipulating planning time and the number of relevant properties could test whether information available in later adjectives changes earlier choices. Speech onset and eye movements could establish whether a redundant property was available before production began or entered consideration during the utterance. Such evidence would connect the fitted balance between complete-description utility and local competition to the time course and resources of planning.

6.2 Semantic Stability during Incremental Production

The context-fixed models predicted production more accurately than models that recomputed the size comparison class during composition. For the participant-specific models, this advantage chiefly concerned adjective content and utterance length; the difference in order conditional on content was small and uncertain. The result makes the maintenance of a scene-based size comparison a plausible component of reference production in this task. A speaker could establish which objects count as large or small during scene apprehension and retain that comparison while selecting adjectives.

The full scene and marked target provide information from which this comparison can be formed before production begins. In discourse, a subset of referents may become relevant through a preceding utterance, potentially changing the context against which size is evaluated. Varying when contextual information becomes available would help identify when speakers retain an initial comparison and when they revise it. Independently varying lexical subjectivity and visual discriminability would further distinguish uncertainty about a property's interpretation from difficulty perceiving it in a particular scene. The current visual manipulation holds lexical subjectivity constant, so its results bear on perceptual reliability within these materials.

6.3 Available Alternatives and Communicative Efficiency

The simulation shows that the availability of shorter descriptions changes the communicative advantage of the same two adjective orders. Under complete-utterance evaluation, updating can generate a size-first advantage by changing the comparison class for size. With prefix evaluation, adding single-adjective alternatives changes the average advantage from colour-first to size-first under both semantic regimes. The value of an order therefore depends on which other descriptions the speaker can choose and which alternatives the listener expects the speaker to have considered. This dependence gives a computational basis for studying ordering within the broader distribution of content and length.

An account of reference production must consequently explain how alternatives become available. Visual attention, lexical accessibility, and discourse can influence which properties enter consideration and how they are combined into candidate descriptions. An utterance-level plan could direct consideration towards properties relevant to the referential goal, while local competition influences which property is mentioned next. Making shorter alternatives more or less available, while holding the scene and two-adjective sequences constant, would test their contribution to interpretation. Comparing production and comprehension under this manipulation would examine whether speakers and listeners rely on corresponding alternative sets.

The predictive findings are conditional on the shared model specification and the response format examined here. The models include stable order expectations, salience, and length and selection terms selected during development with these data; form contributes through salience and the length and selection terms under a neutral literal likelihood. Response-wise leave-one-out prediction evaluates further responses from participants represented in the dataset. Freer production with new participants, materials, and languages would test how the plan-guided account extends beyond these German prenominal descriptions. Together with the remaining predictive discrepancies, these scope conditions define where further evidence can refine the account.

The broader contribution is an account of adjective ordering embedded in the construction of a useful referential description. Scene-based meanings, stable order expectations, and incremental competition jointly shape what speakers choose to mention and how they arrange it. The plan-guided framework connects these choices to the evaluation of a complete description, while the simulation shows how available alternatives affect its interpretation. This combination links adjective ordering to how much speakers plan and how individual choices unfold during communication.

7 Data Availability Statement

Data, analysis scripts, model code, posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the production-model inputs and outputs, documentation of preprocessing and input reconstruction, and the scripts required to reproduce the reported behavioural and predictive analyses.

8 Author Note

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A Supplementary Experimental Methods and Results

A.1 Participants

All studies were conducted online via Prolific, with compensation set slightly above the platform's median level. Participants reported German as their native language.

Study 1a: Slider ratings. Study 1a uses the same dataset reported by Schlotterbeck and Wang (2023). The retained sample included 70 participants in the low- and 64 in the high-discriminability group. Exclusions addressed missing records, extreme completion times, and failures on control or task-relevance checks (Table 3).

Study 1b: Direct replication. The replication used the same task and materials with an independent participant sample. Applying the Study 1a exclusion criteria retained 104 of 132 participants and 2,786 size–colour observations.

Study 2: Production. The retained production sample comprised 60 participants with low size discriminability and 53 with high size discriminability. Participants from either slider study were ineligible. Exclusions addressed extreme completion times, missing or unannotatable responses, and more than 80% identical responses. The latter criterion selects a sample that varies its descriptions across trials.

Table 3 summarises the participant exclusions and the observations included in each analysis. Participants could meet more than one exclusion criterion; the table therefore reports both criterion-specific counts and the total number excluded. For Study 2, the production model comparison reported in Section 4 uses all 9,100 annotated observations across the fifteen utterance categories and three adjective-pair families. The behavioural analyses focus on the size–colour subset.

Table 3: Participant exclusions and observations included in each analysis. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Observations included in the manuscript
Study 1a slider ratings	$150 \rightarrow 134$	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three "none fits" responses on experimental items; 16 unique participants excluded.	All retained critical trials: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	$132 \rightarrow 104$	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	All retained replication trials: 8,379 critical observations. Reported size–colour slider subset: 2,786 observations.
Production study	$149 \rightarrow 113$	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Full production dataset: 9,100 annotated observations. Reported size–colour behavioural subset: 3,034 observations.

A.2 Materials

The noun remained *Aufkleber*, and size, colour, and form varied across counterbalanced lists. Critical scenes were realised through systematic permutations of colour and form values applied to 27 six-object scene templates. This varied the surface displays while preserving the referential structure of each condition.

The slider stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The large-size band comprises $\{9, 10\}$ in both groups; the smaller-size band is $\{1, 2, 3\}$ at high discriminability and $\{1, 2, 3, 4, 5, 6\}$ at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials. The stimulus table specifies the six object sizes, colours, and forms for every list–item–condition combination.

The slider-rating materials used the fixed adjective vocabulary in Table 4. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table 4: Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels `1--6`, `9`, `10`	*großen*, *kleinen*
Colour	`blue`, `orange`, `grey`, `brown`, `green`, `black`	*blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen*
Form	`circle`, `diamond`, `heart`, `quadrat`, `star`, `triangle`	*runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen*

A.3 Design and Hypotheses

The studies shared the reference-game setting and display design described in Section 2. For slider judgements, greater referential usefulness of size predicts stronger size-first preference, yielding the ordering size-sufficient $>$ both-necessary $>$ colour-sufficient; a stable size-first preference predicts ratings above neutrality even when colour alone identifies the target. The slider interaction contrasts assess how discriminability effects differ across contexts, with Study 1b testing whether the pattern observed in Study 1a recurs. For production, the targeted interaction contrasts evaluate whether lower size discriminability increases redundancy more strongly when size alone is sufficient, and whether higher discriminability increases size-initiality more strongly when both properties are necessary. These outcomes connect the selection and ordering of properties to the referential demands of the display.

A.4 Procedure

Each study began with three practice trials with feedback. Main trials were divided into four blocks of approximately equal length, with self-paced breaks and independently randomised trial order within each block. Study 1a participants completed 81 critical trials and 99 filler or control trials (180 total), with a mean completion time of 27.6 minutes. Control trials had an unambiguous expected response; the additional "none fits" option allowed participants to indicate that neither description identified the target. Frequent use of this option on experimental trials was an exclusion criterion, as detailed in Table 3. Study 1b used the same 1,080 combinations of list, item, and condition as Study 1a.

On each critical trial, two phrases differing in adjective order appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation was randomised across trials. Ratings were centred so that zero indicated neutrality, positive values favoured size-first, and negative values favoured colour-first order, on a scale from -0.5 to 0.5.

Participants typed adjective material into the frame *den* ___ *Aufkleber*. Practice trials introduced the relevant colour and form vocabulary. Study 2 participants completed 81 critical and 54 filler or control trials (135 total), with a mean completion time of 32.4 minutes. Fewer filler and control trials were used because typed responses were slower and more effortful than slider judgements. Compensation followed the same rate as Study 1a. Restricting responses to the adjective slot reduced off-task variability and provided a controlled basis for annotation. Annotation rules and the retained response categories are given below.

A.5 Data Analysis

Response coding. The first author annotated production responses using the 15 ordered adjective categories listed in Table 5. The data files use `D` for the size or dimension adjective. The model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 5: Production annotation scheme. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.

Length	Codes
One adjective	`D`, `C`, `F`
Two adjectives	`DC`, `DF`, `CD`, `CF`, `FD`, `FC`
Three adjectives	`DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD`

For modelling, each response is matched to its recorded display. Colour and form are represented by whether each object matches the target on that property; the lexical forms shown to participants are listed above. The categorical response variable for the computational comparison is the ordered utterance label $y_i \in \{1, \ldots, 15\}$. For each binary behavioural outcome, matching responses were coded as 1 and all other annotated size–colour responses as 0.

Behavioural models. The behavioural hypotheses were evaluated with Bayesian hierarchical models. All behavioural analyses used the size–colour subsets: 3,485 observations in Study 1a, 2,786 in Study 1b, and 3,034 in Study 2. The production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete fifteen-category response space and all 9,100 production trials. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \mathrm{Normal}(\mu_{ij}, \sigma), \qquad \mu_{ij} = \mathbf{x}_{ij}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ij}^{\mathsf{T}}\mathbf{b}_i, \qquad \mathbf{b}_i \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{participant}}).$$

For production, size-initiality indicated whether the coded response began with the size adjective. Redundant adjective use, coded as over-informativeness in the analysis, indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k:

$$y_{ijk} \sim \mathrm{Bernoulli}(\pi_{ijk}), \qquad \mathrm{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ijk}^{\mathsf{T}}\mathbf{b}_i + c_k, \qquad c_k \sim \mathrm{Normal}(0, \sigma_{\mathrm{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in the two one-property-sufficient contexts. For redundant adjective use, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\mathrm{Normal}(0, .30)$, $\mathrm{HalfNormal}(.20)$ priors on participant standard deviations, and a $\mathrm{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\mathrm{Normal}(0, 2)$ intercept prior, $\mathrm{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\mathrm{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\mathrm{LKJ}(2)$ prior. Four chains run in parallel each provided $1{,}000$ retained draws after $1{,}000$ warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\widehat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned contrasts were then evaluated on the rating or response-probability scale.

A.6 Detailed behavioural results

Tables 6 and 7 report posterior medians and 95% credible intervals for the behavioural contrasts. Slider estimates are in centred-rating units; production estimates are differences in response probability. The column P reports

the posterior probability of the direction stated in each row; a dash indicates that a directional probability was not separately reported for that contrast in the main analysis summary. Interaction estimates compare effects across contexts, whereas within-context contrasts compare the two discriminability groups in one context.

A.6.1 Slider ratings and replication

Both slider studies supported the predicted referential-context gradient, including a residual size-first preference in colour-sufficient displays (Figure 7). Study 1a also showed a credible interaction in the comparison of size-sufficient and both-necessary displays. Higher discriminability was associated with a shift towards size-first ratings in size-sufficient displays and towards colour-first ratings in both-necessary displays, although the credible intervals for both within-context effects included zero. The interaction therefore concerned the difference between these effects; each individual effect was less precisely estimated. The corresponding interaction between colour-sufficient and size-sufficient displays was less conclusive.

Study 1b did not clearly replicate either interaction contrast. Its estimated effect of higher size discriminability was positive in all three contexts, with intervals above zero in size-sufficient and colour-sufficient displays and an interval spanning zero in both-necessary displays. Thus, clearer size differences could strengthen size-first preference in this replication, with uncertainty about variation in that effect across contexts. The two studies used the same Bayesian hierarchical model specification. Estimates in Table 6 are posterior medians, with contrasts expressed in centred-rating units; directional probabilities refer to the stated contrast under the fitted model.

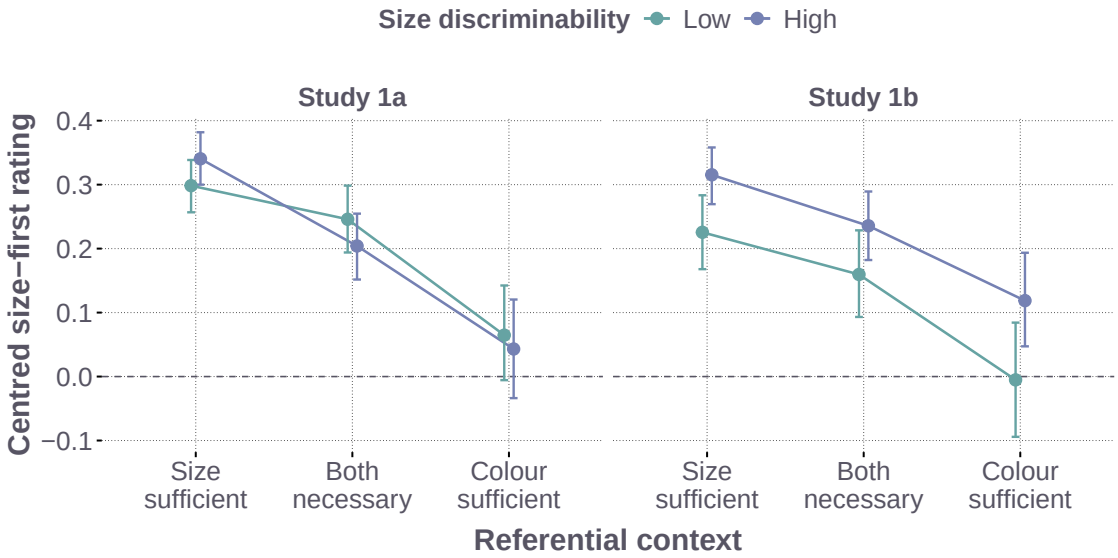


Figure 7: Ordering judgements in Study 1a and Study 1b by referential context and size discriminability. Points show posterior medians of expected centred ratings for a new participant; error bars show 95% credible intervals. Positive values indicate size-first preference; zero indicates neutrality. The two studies use the same model specification and interval definition.

The posterior probability of the graded context order was greater than .999 in Study 1a and .999 in Study 1b. Observed mean centred ratings in Study 1b were .272 in size-sufficient, .197 in both-necessary, and .054 in colour-sufficient contexts. Figure 8 shows the observed Study 1a ratings; Figure 7 shows fitted cell expectations with 95% credible intervals for both slider studies.

A.6.2 Production contrasts

The planned redundancy interaction showed a larger increase under low discriminability in size-sufficient displays than in the other two contexts (Figure 9, Panel A; Table 7). Within size-sufficient displays, lower discriminability increased redundant adjective use; the corresponding increases in both-necessary and colour-sufficient displays were smaller, and their credible intervals included zero.

For size-initiality, the high-minus-low discriminability effect was larger when both properties were necessary than when one property was sufficient (Figure 9, Panel B). Higher discriminability increased size-initial responses

Table 6: Slider-rating contrasts, with interactions followed by context contrasts and within-context discriminability effects. Discriminability effects are high minus low. Interaction rows subtract the size-sufficient discriminability effect from the effect in the named context.

Study	Contrast	Median	95% CrI	P (direction)
1a	Interaction: both-necessary	$-.085$	$[-.151, -.022]$	.995 (< 0)
1a	Interaction: colour-sufficient	$-.063$	$[-.172, .039]$	.885 (< 0)
1a	Size-sufficient minus both-necessary	.094	$[.061, .127]$	—
1a	Both-necessary minus colour-sufficient	.171	$[.131, .208]$	—
1a	Colour-sufficient rating relative to zero	.054	$[.003, .109]$	.982 (> 0)
1a	Discriminability: size-sufficient	.042	$[-.013, .099]$	—
1a	Discriminability: both-necessary	$-.043$	$[-.118, .028]$	—
1a	Discriminability: colour-sufficient	$-.021$	$[-.129, .082]$	—
1b	Interaction: both-necessary	$-.014$	$[-.097, .071]$	.623 (< 0)
1b	Interaction: colour-sufficient	.034	$[-.082, .157]$	.285 (< 0)
1b	Size-sufficient minus both-necessary	.073	$[.029, .115]$	—
1b	Both-necessary minus colour-sufficient	.140	$[.090, .190]$	—
1b	Colour-sufficient rating relative to zero	.057	$[.001, .116]$	.977 (> 0)
1b	Discriminability: size-sufficient	.090	$[.014, .160]$	—
1b	Discriminability: both-necessary	.076	$[-.011, .161]$	—
1b	Discriminability: colour-sufficient	.124	$[.016, .240]$	—

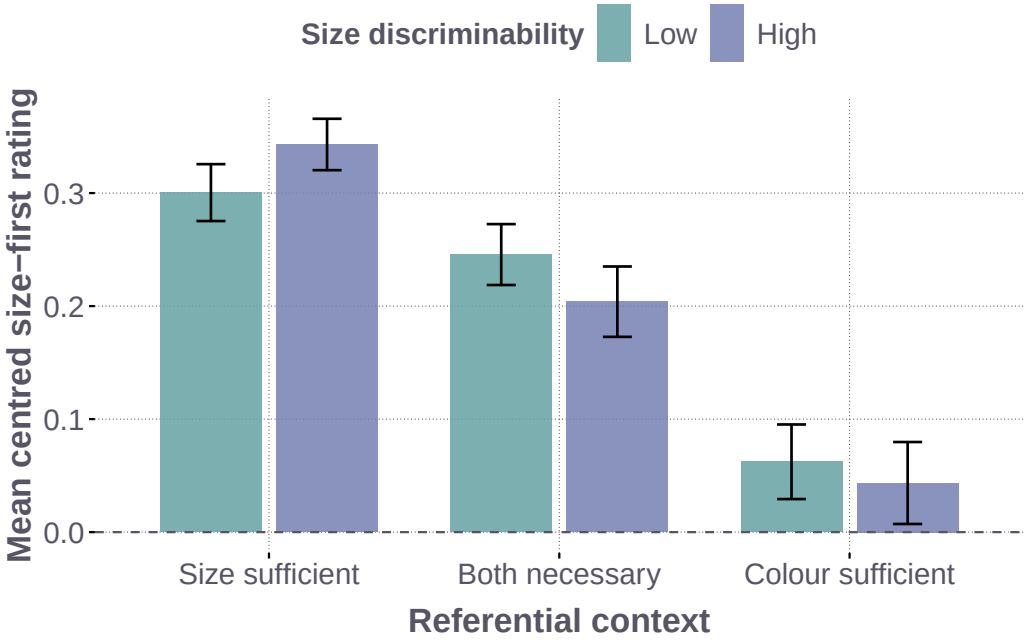


Figure 8: Observed Study 1a slider ratings. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

in both-necessary displays. The effect in size-sufficient displays was close to zero and uncertain, whereas colour-sufficient displays showed a small decrease. Averaged across discriminability conditions, size-initial responses were more probable in size-sufficient than in both-necessary displays, and more probable in both-necessary than in colour-sufficient displays. Size-initiality includes descriptions consisting of size alone. Among responses containing both size and colour, size-first descriptions remained more frequent, locating context sensitivity in adjective selection, length, and initial position alongside this conditional order preference.

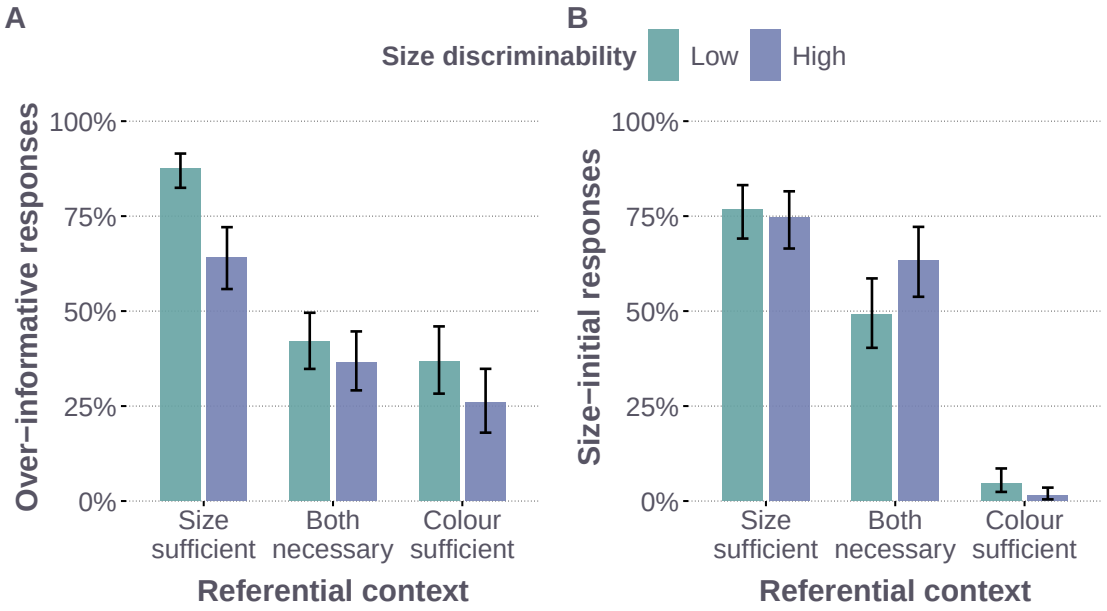


Figure 9: Redundant adjective use and size-initial responses in size–colour production. Panel A shows redundant adjective use: a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel B shows the probability that a response begins with size. Bars show Bayesian posterior medians and error bars show 95% credible intervals. Size-initiality combines adjective selection and order.

Table 7: Production contrasts on the response-probability scale. The redundancy interaction compares the low-minus-high difference in size-sufficient displays with its mean in the other two contexts. The size-initiality interaction compares the high-minus-low difference in both-necessary displays with its mean in the two one-property-sufficient contexts.

Contrast	Median	95% CrI	P (direction)
Redundancy: planned interaction	.153	[.072, .236]	$>$ .999 ($>$ 0)
Redundancy: size-sufficient, low minus high	.234	[.151, .318]	$>$ .999 ($>$ 0)
Size-initiality: planned interaction	.169	[.077, .256]	$>$ .999 ($>$ 0)
Size-initiality: both-necessary, high minus low	.141	[.036, .242]	.997 ($>$ 0)
Size-initiality: size-sufficient, high minus low	−.021	[−.101, .057]	—
Size-initiality: colour-sufficient, high minus low	−.032	[−.068, −.006]	—
Size-initiality: size-sufficient minus both-necessary	.193	[.139, .248]	—
Size-initiality: both-necessary minus colour-sufficient	.531	[.459, .601]	—

Observed redundancy proportions in size-sufficient displays were .90 under low and .63 under high size discriminability. Observed size-initial proportions in both-necessary displays were .49 under low and .65 under high discriminability. These descriptive proportions differ from the posterior-predictive contrasts in Table 7, which integrate over participant and item variation.

A descriptive check of the complete production data also showed group differences in the colour–form family. In form-sufficient displays, form-initial responses were more frequent in the high- than in the low-discriminability group (.77 vs. .71). Because discriminability was manipulated between participants, these differences can also reflect group variation in how the task was approached. The production model represents participant variation alongside the visual manipulation.

B Production Model Specification

B.1 Representation of Experimental Displays

Participant list, condition, and item identifiers link each retained response to its recorded six-object display. The model represents each display as a 6×3 array of size, colour, and form values, with the intended referent indexed first. This representation preserves the observed values of all three properties, including those outside the adjective pair defining the trial.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the comparison class $\mathcal{C}$ (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta\left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C})\right), \tag{8}$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o, graded membership is

$$m_{\mathrm{S}}(o \mid \mathcal{C}) = \Phi\left(\frac{s_o - \theta_\delta(\mathcal{C})}{w\sqrt{s_o^2 + \theta_\delta(\mathcal{C})^2 + \epsilon}}\right), \qquad w = .6856, \quad \epsilon = 10^{-8}, \tag{9}$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1 - \nu$, with $\nu_{\mathrm{C}} = .59$ and $\nu_{\mathrm{F}} = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (2), (3), (5) and (6) define how this meaning enters the literal listener L and the three production accounts.

Figure 10 separates fixed theoretical choices, estimated parameters, participant variation, and observed responses. The fixed variables A and R index production architecture and semantic regime. For participant i on trial t, the model maps the display and target X_{it} and the stable-order score $\Omega(u)$ to a probability $\pi_{it}(u)$ over the fifteen possible utterances, from which the observed response y_{it} is drawn. The shared coefficients govern visual salience, adjective selection, and utterance length.

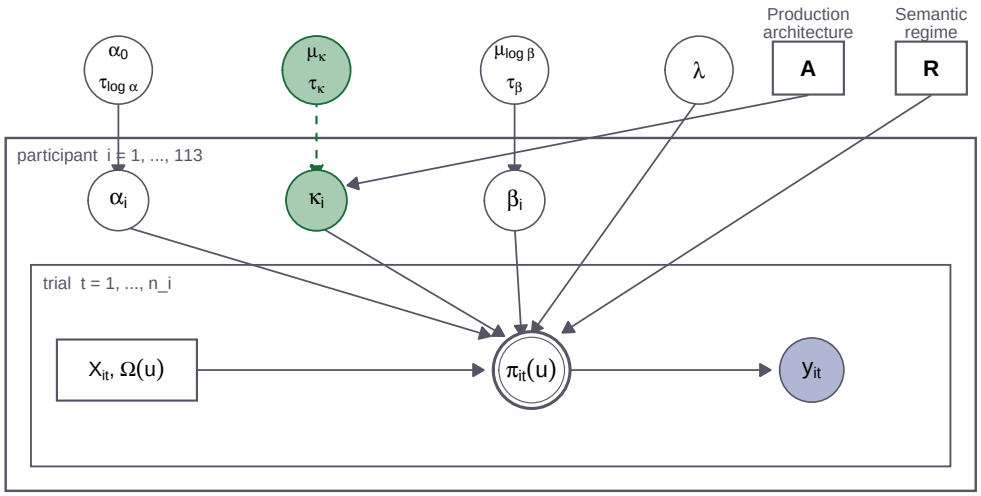


Figure 10: Hierarchical structure of the production model and the restrictions used in the matched comparisons. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The dashed κ hierarchy applies to the plan-guided model with participant-specific incremental-choice weights.

B.3 Component Inventory

Table 8: Components of the production models.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Graded support for matching colour
Form reliability	Fixed neutral	Form production supported by the length and selection terms
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Length and selection coefficients	Estimated	Support for sufficient single adjectives, form use, and length and discriminability effects
Incremental-choice weight κ_i	Estimated in plan-guided production	Shared weight or participant-specific weights for incremental adjective choice

The population pragmatic-optimality parameter has prior $\alpha_0 \sim \mathrm{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log\beta} \sim \mathrm{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log\alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log\beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log\alpha} \sim \mathrm{HalfNormal}(.15)$, and $\tau_\beta \sim \mathrm{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \mathrm{Uniform}(0, 1)$. In the model with participant-specific incremental-choice weights, $\mu_\kappa \sim \mathrm{Uniform}(0, 1)$ and $\kappa_i = \mathrm{logit}^{-1}(\mathrm{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \mathrm{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \mathrm{Normal}(0, 1)$. The salience coefficient has a $\mathrm{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\mathrm{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\mathrm{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\mathrm{HalfNormal}(1)$ prior. Threshold selectivity is fixed at $\delta = .50$; colour reliability, form reliability, and perceptual blur take the values specified above. A final Laplace smoothing step mixes the predicted distribution with a uniform distribution over the fifteen responses, with a fixed weight of .003 on the uniform component.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the displayed scene. For the target o^*, let s_o, c_o, and f_o be an object's recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_{\mathrm{S}} = \frac{1+h}{2}\ \mathrm{logistic}\left(\frac{s_{o^*} - \max_{o\neq o^*} s_o}{\mathrm{sd}(s) + 10^{-8}}\right), \tag{10}$$

$$r_{\mathrm{C}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \qquad r_{\mathrm{F}} = 1 - \frac{1}{5}\sum_{o\neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \tag{11}$$

Here $\mathrm{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_{\mathrm{S}}, b_{\mathrm{C}}, b_{\mathrm{F}}) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3}\sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_{\mathrm{salience}} z_a$ and the continuation cost in the length and selection terms below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u, the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|}\sum_{v\in\mathcal{U}_{A(u)}} \log q(v). \tag{12}$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Length and selection terms. The length and selection terms $T(u, X)$ connect the referential design to choices about which properties to mention and how far to continue a description. Let $n = |u|$ and $A(u)$ denote the length and unordered adjective set of u. For this term, X includes the display's condition labels: $d \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}, \varnothing\}$ is the property that identifies the target on its own, with $d = \varnothing$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the production models use

$$\begin{aligned} T(u, X) = & -\rho_{\text{cont}} \sum_{t=1}^{n-1} \max(z_{w_t}, 0) + \lambda_{\text{suff}} \mathbf{1}\{d \neq \varnothing,\ u = (d)\} \\ & + \lambda_{\mathrm{F}}(1 - c)\mathbf{1}\{\mathrm{F} \in A(u)\} - \lambda_3 \mathbf{1}\{n = 3\} \\ & + \lambda_{\mathrm{S}} h\, \mathbf{1}\{d = \mathrm{S},\ u = (\mathrm{S})\} \\ & + \lambda_{\mathrm{SF}}(1 - 2h)\mathbf{1}\{d = \mathrm{S},\ A(u) = \{\mathrm{S}, \mathrm{F}\}\}. \end{aligned} \tag{13}$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (3). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing descriptions outside colour-sufficient conditions. With form reliability fixed at .50, form gives equal literal support to matching and nonmatching objects; this task term provides a direct contribution to form use. The three-adjective penalty offsets the value accumulated along longer descriptions. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Numerical evaluation and response likelihood. The implementation uses 64-bit arithmetic and a floor of 10^{-8} for size memberships, listener target probabilities before taking logarithms, and local choice probabilities. Colour and form memberships receive an additive 10^{-8} offset. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of floored local log-choice probabilities, and $B(u) = \beta_i \Omega(u) + T(u, X)$, the computed score before Laplace smoothing is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\widetilde{S}_{\kappa_i}(u \mid X)$, whose endpoints obey the geometric interpolation in Equation (18). The categorical response likelihood is

$$p(y_{it} = u \mid X_{it}) = (1 - .003)\widetilde{S}_{\kappa_i}(u \mid X_{it}) + \frac{.003}{15}. \tag{14}$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ; the model with participant-specific incremental-choice weights estimates κ_i. All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

C Production Model Comparison and Diagnostics

C.1 Inference and Predictive Comparison

Each likelihood evaluation scores all fifteen completed utterances and performs the required semantic composition and prefix-level normalisation. To make this exact calculation feasible, we precomputed trial-independent utterance and prefix structures and vectorised and compiled the listener and speaker computations in JAX. All eight production fits used four No-U-Turn Sampler (NUTS) chains on a GPU and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8 to obtain adequate mixing; the other six models retained 1,000 draws per chain at depth 6. The eight fits had zero divergences, maximum $\widehat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum Bayesian fraction of missing information (BFMI) above .64, and no observation with Pareto $k > .7$. The participant-weight plan-guided fits reached

the depth-6 step cap on 77.7% of transitions under fixed semantics and 31.7% under updating; these frequent depth-limit hits indicate restricted sampling efficiency, alongside the convergence and effective-sample-size values reported above. The best-predicting context-fixed model had an absolute ELPD of $-13{,}039.924$, maximum $\widehat{R} = 1.0051$, and minimum BFMI of .742.

Implementation checks verified that each modifier is applied once from the noun outwards and that changing the semantic regime can change predictions at matched parameters. Endpoint, interpolation, and direct-versus-precomputed probability checks agreed within numerical precision, and recomputed posterior log likelihoods agreed with the stored values within 10^{-8}.

The pointwise leave-one-out score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is another response from a participant represented in the dataset, as described in Section 4.2.

Table 9: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy.

Contrast	ΔELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ, minus average endpoints	402.677	$[311.466, 503.098]$	$> .999$
Fully incremental minus global	-291.596	$[-684.673, -1.321]$	$.024$
Updating minus fixed, shared-κ grid	-43.527	$[-56.005, -31.770]$	$< .001$
Participant κ_i hierarchy, context-fixed	364.742	$[248.178, 520.967]$	$> .999$
Participant κ_i hierarchy, updating	372.285	$[255.597, 526.056]$	$> .999$

Averaging over the two semantic regimes, the plan-guided model had the highest predictive accuracy of the three architectures in .994 of the Bayesian-bootstrap draws. In the context-fixed plan-guided model, the posterior mean shared incremental-choice weight was

$$\kappa = .446, \qquad 95\% \text{ credible interval} = [.407, .486]. \tag{15}$$

C.2 Architecture-Level Response Patterns

Figure 11 shows initial-adjective cells in which the largest difference among architecture predictions is at least 6.7 percentage points and traces these differences into completed utterances. Colour-initial responses contributed 328.912 ELPD to the matched plan-guided model's advantage over the average endpoint, or 81.7% of the summed pointwise contrast. Across the displayed cells, plan-guided production often lies between the global and fully incremental endpoints. The response partitions locate this gain by summing exact-response log scores within the observed categories.

C.3 Decomposing the Semantic Predictive Contrast

The architecture-specific updating-minus-fixed effects were -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible interval excluded zero. The sequentially updating participant-specific model was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$.

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-it}(u \mid X_{it}) = p_{-it}(\ell(u) \mid X_{it})\, p_{-it}(A(u) \mid \ell(u), X_{it})\, p_{-it}(u \mid A(u), X_{it}). \tag{16}$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. All three contributions refer to the same held-out response. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws using seed 20260906. For the participant-weight plan-guided pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the participant-weight pair, the original response contrast sums to -17.548 ELPD in size–colour trials, -51.172 in size–form trials, and 14.580 in colour–form trials. These partitions locate the predictive difference within the complete response distribution.

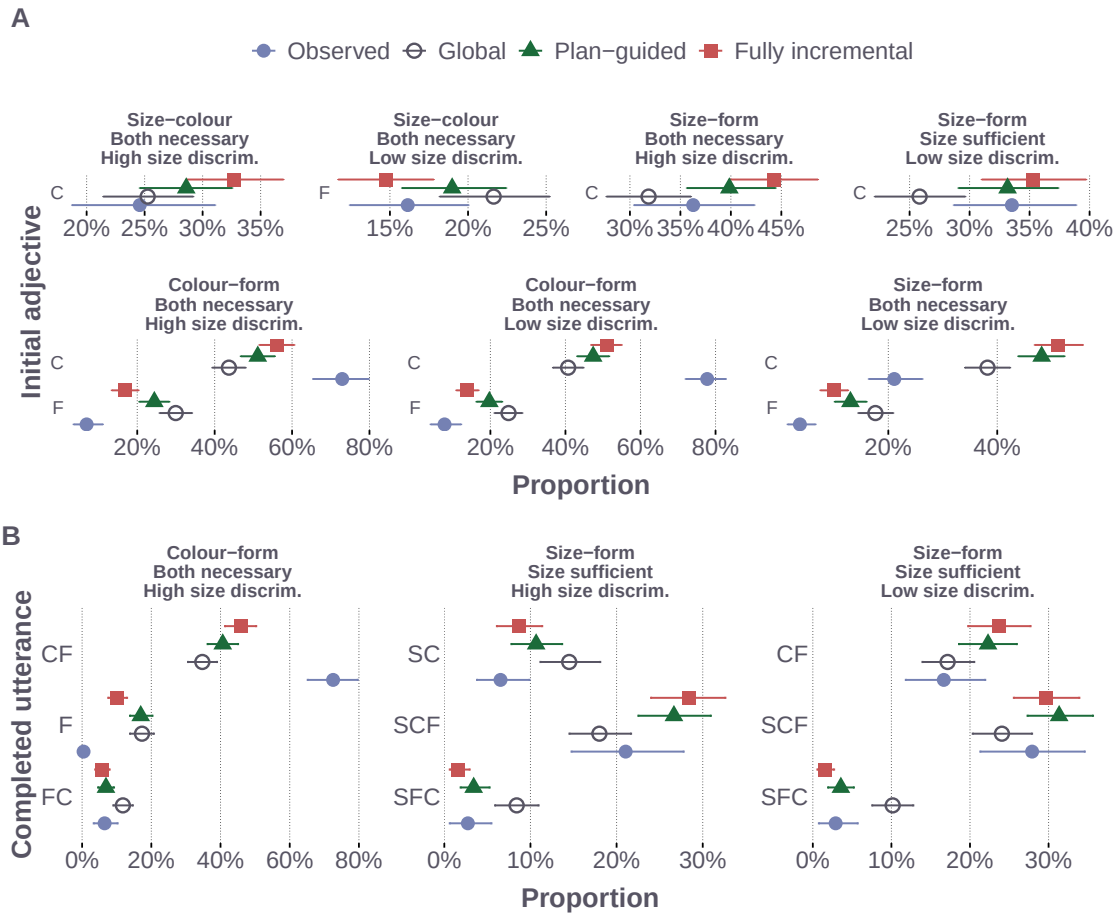


Figure 11: Qualitative comparison of the matched production architectures. Panel A shows initial-adjective cells for which the maximum difference among model predictions is at least 6.7 percentage points. Panel B shows completed utterances with differences of at least five percentage points within the three experimental conditions with the largest pairwise total-variation distance among model predictions. Response labels use S for size, C for colour, and F for form. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Horizontal scales vary across conditions to make the comparatively small differences among production architectures visible. The displayed cells are descriptive diagnostics; the held-out comparison in Figure 4 provides the evidence for relative predictive accuracy.

Response residuals. We compared the two participant-weight plan-guided models using the posterior mean probability of each utterance in each combination of adjective family, referential context, and size discriminability. For each of the 270 response cells, the residual was the observed response proportion minus the mean predicted probability, expressed in percentage points. Positive residuals indicate underprediction and negative residuals indicate overprediction. These descriptive residuals use the full-data posterior; the ELPD decomposition above evaluates held-out responses. Across cells, mean absolute residuals were 3.04 percentage points for context-fixed semantics and 3.08 for updating; root mean squared residuals were 6.03 and 6.07, respectively. Table 10 shows the six cells with the largest absolute between-model changes in absolute residual error, retaining changes that favoured either model. All six occurred in size-sufficient displays. In size-sufficient size–form trials with high discriminability, both models overpredicted three-adjective size–colour–form responses: the observed proportion was 21.1%, compared with 27.3% under fixed semantics and 31.2% under updating. This response showed the largest change in absolute residual error between the models. In the same condition, both models underpredicted size–form utterances, with a larger discrepancy under updating. Updating reduced errors for size-only responses in both size-containing families under high discriminability. The held-out score partition likewise located the largest net disadvantage of updating in the size–form family, as reported above.

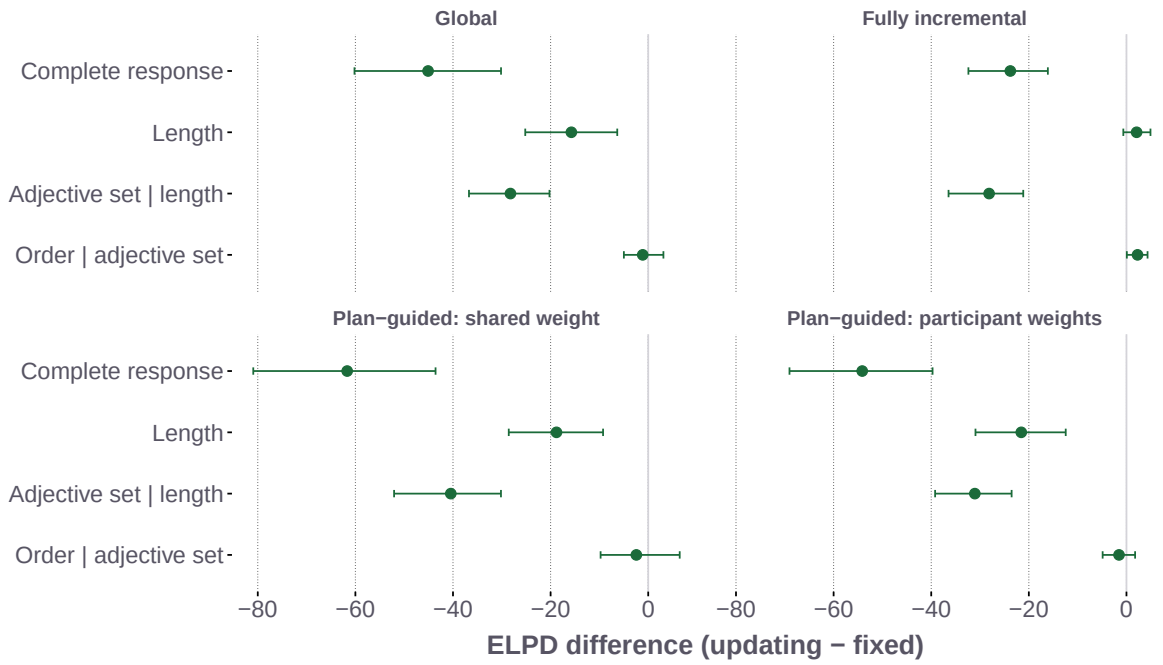


Figure 12: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific incremental-choice weights; the other plan-guided panel uses a shared weight.

Table 10: Largest changes in response residuals between the participant-weight plan-guided semantic models. All rows concern size-sufficient displays. Observed proportions are percentages; residuals are observed minus predicted proportions in percentage points. S, C, and F denote size, colour, and form, with surface order preserved.

Adjective family	Discriminability	Utterance	Observed	Fixed residual	Updating residual
Size–form	High	S–C–F	21.1	−6.2	−10.1
Size–colour	High	S	27.2	2.7	0.9
Size–form	High	S	25.5	−2.3	−0.6
Size–form	Low	S–C–F	27.9	−4.1	−5.7
Size–form	High	S–F	21.5	12.9	14.4
Size–form	Low	S–F	19.8	8.8	9.8

C.4 Posterior-Predictive Adequacy

The condition-wise response distributions appear in Figure 13. Across the 270 cells formed by the fifteen completed utterances, three adjective families, three referential contexts, and two size-discriminability conditions, the posterior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points. The corresponding 95% posterior-predictive intervals contained 58.5% of the observed cell proportions, a descriptive check of fit across the dependent response cells.

In size–colour trials, the model reproduced the predominance of colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. It nevertheless overpredicted colour-only responses in colour-sufficient conditions. In size-sufficient and both-necessary conditions, it tended to overpredict size–form responses and underpredict size–colour–form responses. These discrepancies indicate that the model captured the broad changes in descriptive content while misestimating the balance between some two- and three-adjective utterances.

A further discrepancy occurred in size–form trials when both properties were necessary and size discriminability was high. The model underpredicted size-initial responses (predicted: 45.2%; observed: 59.5%) and overpredicted form-initial responses (predicted: 13.5%; observed: 4.2%). The checks therefore identified remain-

ing errors in initial-adjective choice as well as in the distribution of completed utterances.

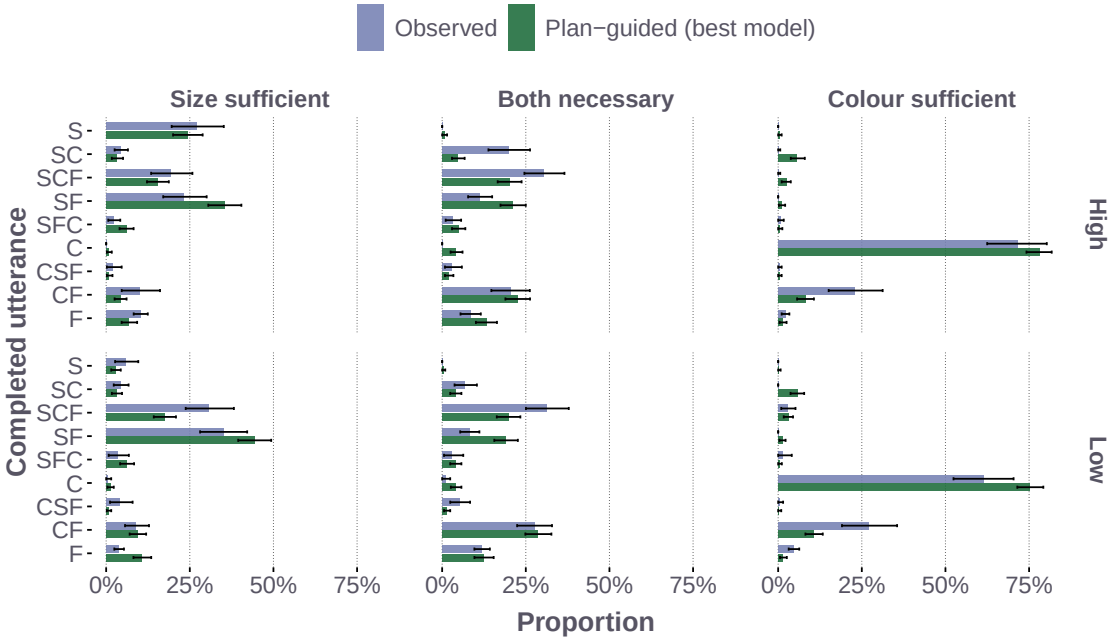


Figure 13: Observed and posterior-predicted proportions for frequent utterances in size–colour trials under the best-predicting plan-guided model. The figure includes utterances whose observed or mean posterior-predicted proportion reaches 5% in at least one context-by-discriminability condition, with proportions pooled across trials within each condition. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Adjective labels preserve surface order: S denotes size, C colour, and F form.

D Model-Development Checks

Model development used the present data to select the form semantics, stable-order feature, and length and selection terms. The matched comparisons hold these components constant across production architectures and semantic regimes.

The reported models fix form reliability at $.50$ and include a coefficient for form use, the GPT-2 order residual, sufficient-single-adjective support, a three-adjective penalty, and size-discriminability terms. Exploratory variants used informative form meanings tied to colour reliability, alternative priors on the form-use coefficient, an additional reliability-based order term, or no stable-order score. A reduced specification also removed the length and selection terms as a bundle. These variants were evaluated before the semantic implementation was corrected and were not refitted. They informed model development; the predictive comparisons reported here use the common specification in Section B.

E Participant Variation and Predictive Contributions

The joint model estimates pragmatic optimality, incremental-choice weight, and stable-order weight for each participant. Table 9 reports its predictive gain relative to the otherwise matched shared-κ model under each semantic regime. In the best-predicting context-fixed model, the population location of the participant-specific incremental-choice weights was $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$.

Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of $[.525, .763]$ and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval

$[163.012, 319.334]$, and 183.796 ELPD on three-adjective utterances, 95% credible interval $[100.757, 304.826]$. Its contribution to one-adjective utterances was -48.426 ELPD (95% credible interval $[-70.371, -28.666]$).

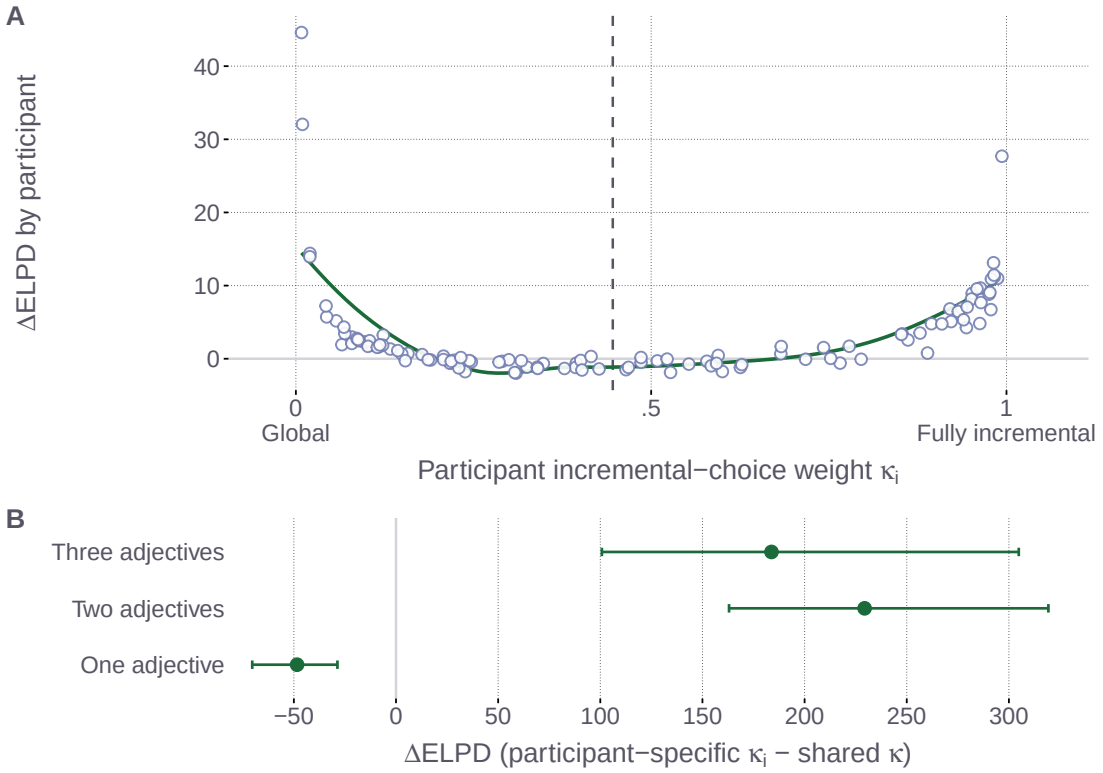


Figure 14: Where participant-specific incremental-choice weights improve prediction. Panel A relates each participant's leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i; the dashed line marks the shared-κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-predicting plan-guided model and its otherwise matched shared-κ model to localise the held-out model contrast.

Posterior-predictive contrasts between participants in the lower and upper tails of each parameter describe the associated differences in utterance choice. Higher α_i concentrates probability on a sufficient single adjective, especially S in size-sufficient displays and F in form-sufficient displays. Higher κ_i mainly redistributes probability among form-involving responses whose continuation sets differ after the first adjective. Higher β_i shifts probability from single-adjective responses towards longer responses that follow the stable order score. In the context-fixed joint model, the mean participant-level posterior correlations were .010 for α_i with κ_i and .025 for α_i with β_i. The posterior participant-level association between κ_i and β_i was positive, with mean $r = .162$ and 95% credible interval $[.073, .249]$. These correlations and tail contrasts describe the associations between participant parameters and response probabilities within the joint fit.

Three-adjective utterances accounted for 1,633 of the 9,100 modelled responses, or 17.9%. Within the fitted model, these responses balance accumulated referential value and salience against a shared three-adjective cost, with α_i, κ_i, and β_i determining how that balance is expressed for each participant. These quantities jointly determine the predicted frequency and ordering of three-adjective utterances.

F Local Normalisation and Model Endpoints

Local adjective competition. For the utilities defined in Equation (3), taking the logarithm of the product of locally normalised choices gives the accumulated utility minus the local-normalisation term. Suppressing the

shared referent and scene arguments,

$$\log \prod_t \frac{\exp U_t(w_t \mid u_{<t})}{\sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t})} = \sum_t U_t(w_t \mid u_{<t}) - \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}) \\ = U^{\text{complete}}(u) - \mathcal{N}^{\text{local}}(u). \tag{17}$$

Relation between the endpoints. At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The stable-order score, length and selection terms, and utterance space are identical across the three architectures. Numerical checks found agreement in category probabilities and log probabilities within a tolerance of 10^{-10}.

Plan-guided interpolation. Before the final Laplace smoothing, the same model can be written at common parameter values as a geometric interpolation between the global and fully incremental endpoints:

$$S_\kappa(u \mid o^*, X) \propto S_{\text{G}}(u \mid o^*, X)^{1-\kappa} S_{\text{I}}(u \mid o^*, X)^{\kappa}, \tag{18}$$

where S_{G} and S_{I} are the endpoint distributions evaluated with the same listener, stable-order component, length and selection terms, and parameter values. The proportionality in Equation (18) renormalises the interpolated scores over the fifteen completed utterances.

G Sequential Context Updating for Size

For the hierarchical semantic path, let C_{j-1} be the context state after the syntactically inner modifiers have composed. The semantic regime determines the comparison class used for size:

$$[\![\text{S}]\!]_{\text{fix}}(o) = [\![\text{S}]\!]_{\theta_\delta(\mathcal{C})}(o), \tag{19a}$$

$$[\![\text{S}]\!]_{\text{upd},C_{j-1}}(o) = [\![\text{S}]\!]_{\theta_\delta(C_{j-1})}(o). \tag{19b}$$

where θ_δ derives a graded size threshold from a comparison class and $\mathcal{C}$ is the original scene, represented by the uniform prior P_0. The context-updating listener recomputes the size threshold from the hierarchically prior state; the context-fixed listener anchors it to the scene prior. At each surface position, the production calculation evaluates the current prefix using this noun-outward interpretation. Each prefix is interpreted from the original object prior, with each modifier applied once.

H Simulation Parameter Sweeps

The original two-order simulation compares a global rule based on completed-utterance utility with an incremental rule that includes continuation-versus-stop competition. We report its scene construction, definitions, and parameter sweeps first. Section H.1 then introduces the production specification on the same scenes, as in the controlled comparison in Section 5.

Each random display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to $[1, 30]$. The largest object is designated as the target and assigned blue; each non-target object is independently assigned blue or non-blue with equal probability. Three levels of contextual size spread vary how distinctly the largest target differs from its competitors. The semantic grid varies colour reliability, size-threshold selectivity, and perceptual blur, which control the strength of colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. In the simulation notation, k corresponds to threshold selectivity δ and w_f to perceptual blur w in Equations (8) and (9).

Figures 16 to 18 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, X)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b, denote the original global and incremental rules by S_G and S_I:

$$S_G(ab \mid o, X) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \tag{20}$$

$$R_I(ab \mid o, X) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \tag{21}$$

$$S_I(ab \mid o, X) = \frac{R_I(ab \mid o, X)}{R_I(ab \mid o, X) + R_I(ba \mid o, X)}. \tag{22}$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The original global rule supplies the complete-utterance evaluation used in the first row of Panel A in the main comparison. The production global model accumulates utility across adjective positions; its incremental endpoint assigns utterance length through the completed-response distribution. The pragmatic listener uses a uniform referent prior, giving $L^{\text{prag}}(o \mid ab, X) = S(ab \mid o, X) / \sum_r S(ab \mid r, X)$. The implementation clips very small intermediate probabilities at 10^{-20}.

The reproducible sweep uses seed 20260906 and samples 1,000 independent scenes for each context-size and size-spread combination. Each scene is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent scenes and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within scenes; repeated evaluations of one scene do not count as independent samples. The computations use CUDA and 64-bit arithmetic.

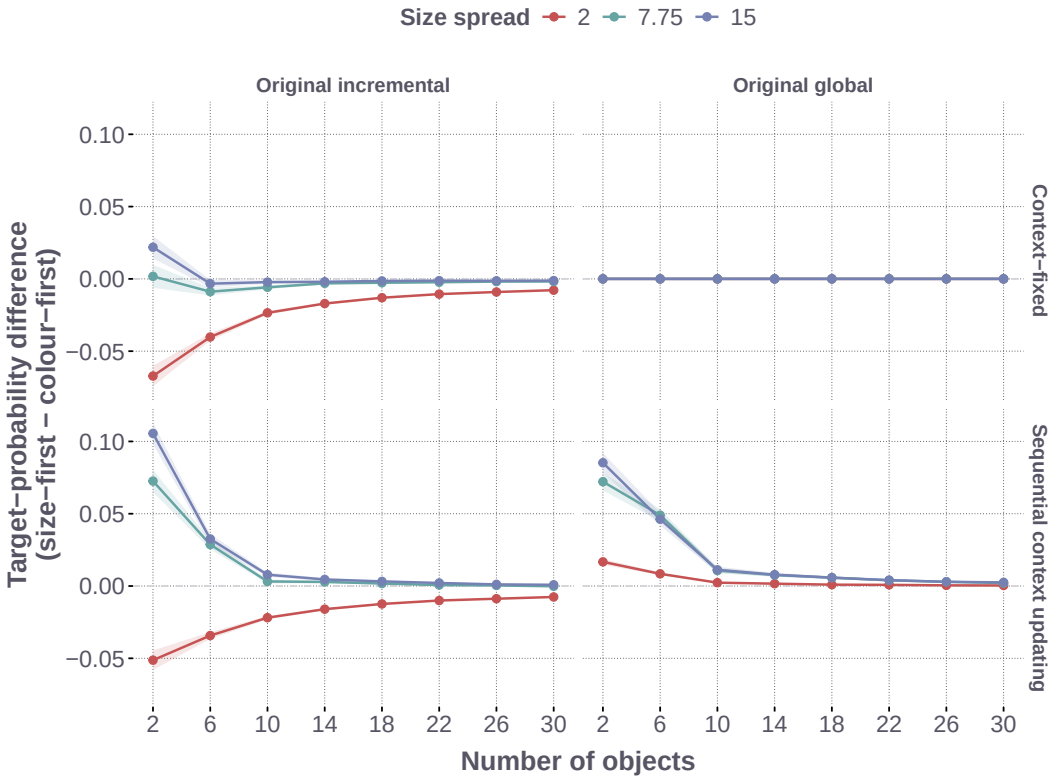


Figure 15: Size-first advantage Δ_{SF} across context size and contextual spread, shown for the original two-order global and incremental rules under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled scene. The complete parameter grid appears in Table 11. Positive values indicate a higher model-assigned target probability for the size-first order.

Figure 15 shows how the size-first advantage changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Wider size distributions generally make the largest target more distinguishable from its competitors; clustered sizes reduce this distinction.

The original global rule with context-fixed semantics provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first advantage under global production. The advantage is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

The original incremental rule adds a contribution from local informativity along the surface sequence. With closely clustered sizes, this rule yields a colour-first advantage in pragmatic target recovery. As sizes spread apart, the colour-first advantage weakens. After averaging over the semantic parameter settings in Figure 15, the original incremental rule yields colour-first advantages in some contexts; the original global rule remains symmetric under

fixed semantics and yields a size-first advantage under updating. At individual parameter settings, the original global rule with updating can also yield small negative estimated advantages. Under the original incremental rule, sequential context updating increases the size-first target-recovery advantage when the colour-restricted comparison class changes size interpretation sufficiently strongly. The parameter sweeps in Figures 16 to 18 show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

Table 11: Simulation parameter grid

Parameter	**Grid values**	**# levels**
Original two-order rule	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
Colour reliability ν	0.90, 0.92, 0.94, 0.96, 0.98	5
Threshold selectivity k	0.10, 0.30, 0.50, 0.70, 0.90	5
Perceptual blur w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Scenes per context-size/spread pair	1,000	
Total model evaluations	**12,000,000**	

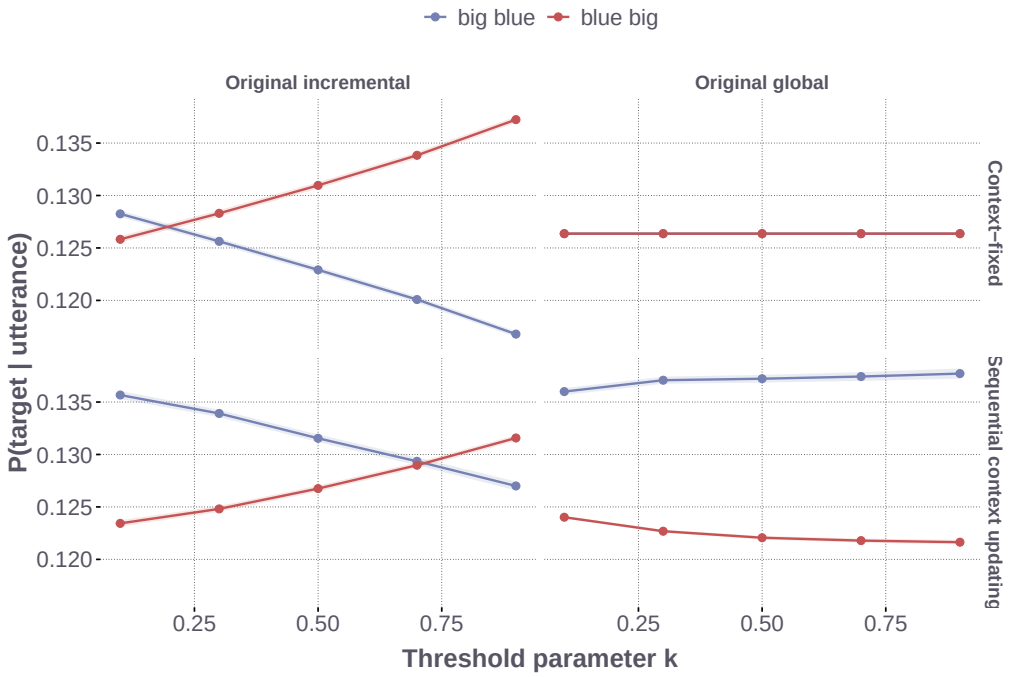


Figure 16: Effect of the threshold parameter k on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

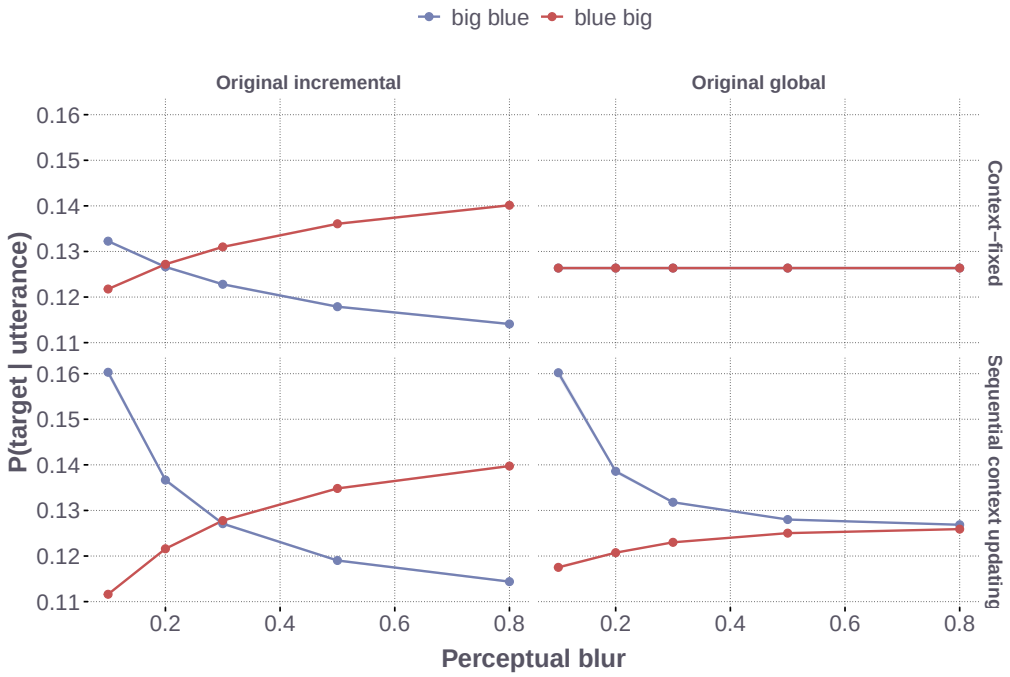


Figure 17: Effect of perceptual blur w_f on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

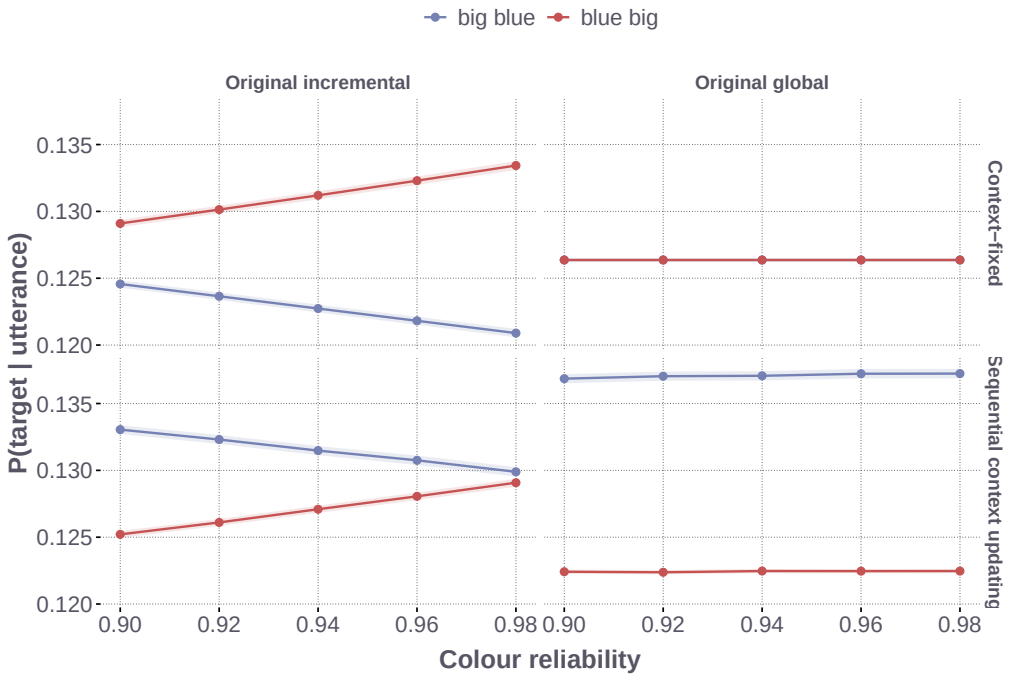


Figure 18: Effect of colour reliability ν on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

H.1 Controlled Changes on the Same Random Scenes

Complete-utterance evaluation, labelled terminal evaluation below, uses a different utility from prefix evaluation:

$$U_{\text{term}}(u, o, X) = \alpha \log L(o \mid u, X), \tag{23}$$

$$U_{\text{prefix}}(u, o, X) = \alpha \sum_{t=1}^{|u|} \log L(o \mid u_{1:t}, X). \tag{24}$$

Both scores are globally normalised over completed utterances. The prefix score with $\kappa = 0$ is the global model used in the main text; $\kappa = .5$ and $\kappa = 1$ give the plan-guided and fully incremental simulation models. The terminal score supplies a control for the additional contribution of intermediate adjective information.

The controlled comparisons reuse all 24,000 scenes from the original sweep and the target-recovery measure in Equation (7). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order's probability under the full speaker distribution, without first conditioning on the two orders. The comparisons include terminal evaluation and prefix evaluation at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. The full fifteen-response implementation agrees with the production forward calculation with task, salience, and order terms set to zero and final Laplace smoothing omitted. This isolates the computational components without assigning fitted participant effects or experimental task labels to random scenes.

The main comparison holds pragmatic optimality at $\alpha = 1$ and excludes stable-order expectations, salience, length and selection terms, and final Laplace smoothing. Form reliability is fixed at $.50$. Panel A of the main figure separates the effects of evaluating information at each adjective position and expanding the utterance space, while holding global choice fixed. Panel B compares global, plan-guided, and fully incremental production using $\kappa = 0, .5, 1$, respectively, at the production semantic constants. The production constants are colour reliability $.59$, threshold selectivity $.50$, and perceptual blur $.6856$. Table 12 reports the mean advantages in both panels of Figure 6. The prefix-evaluation rows specify the architecture through κ; terminal evaluation uses global choice. For the fifteen-response global model at the production constants, the paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$.

Table 12: Mean size-first pragmatic target-recovery advantage in percentage points on the same 24,000 random scenes.

Evaluation	Responses	Semantic settings	Context-fixed	Updating
Terminal	2	Parameter grid	0.000	1.141
Prefix ($\kappa = 0$)	2	Parameter grid	-2.718	-1.838
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962

The main figure uses the production semantic implementation throughout. As a separate implementation control, substituting its lexical calculation into the original two-order global rule reduces the mean updating advantage from 1.465 to 1.142 percentage points across the original grid. The terminal control additionally retains the production listener log-probability floor, giving the 1.141-point advantage in Table 12. The implementations differ in probability floors and numerical normalisation around weighted quantiles; the control measures their combined impact. The original curves are retained in Figure 15.

Figure 19 reports the fifteen-response comparison across evaluation rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This representative setting comes from the fixed model's population log-weight location; it does not integrate posterior or participant uncertainty. At $\kappa = 0$, adding this term gives mean advantages of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive advantages are already present without this term. All other salience, length, and selection terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each scene before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups. These averages summarise the specified parameter grid; individual contexts can favour different orders. Reproduction checks

recovered the stored two-order outputs, and the controlled probabilities agreed with the corresponding production calculation to numerical precision.

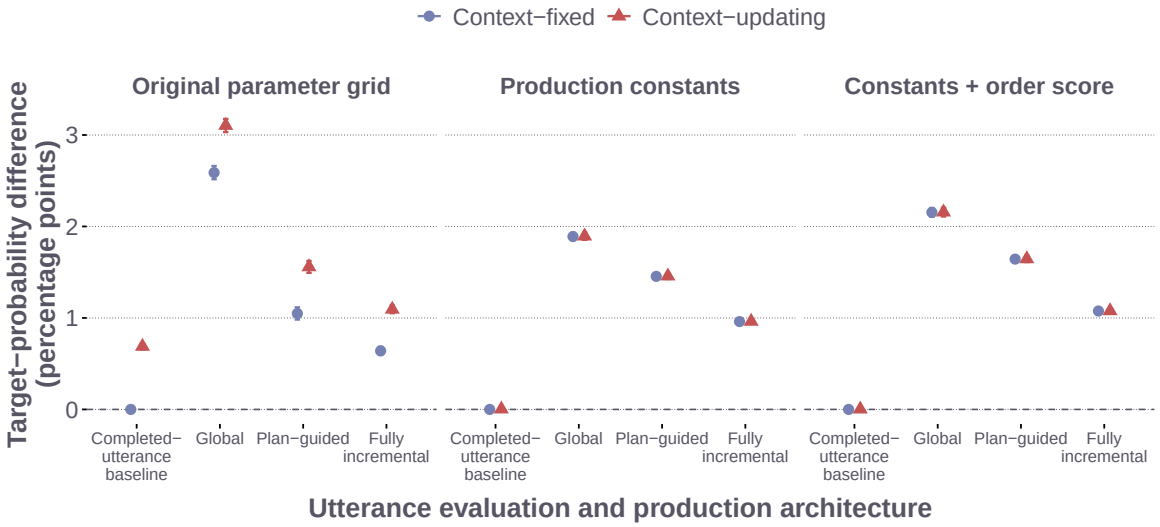


Figure 19: Controlled fifteen-response comparison across evaluation rules, semantic parameters, and a shared stable-order score. The completed-utterance baseline uses only the final literal-listener target probability. The production models accumulate prefix scores: $\kappa = 0$ gives global production, $\kappa = .5$ plan-guided production, and $\kappa = 1$ fully incremental production. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 scene-level Monte Carlo standard errors. All comparisons use the same random scenes; no participant posterior uncertainty is represented.

H.2 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the 3,034 size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and length and selection terms. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote. Because the production responses were coded by adjective type, fixing these named adjective extensions across candidate referents is an additional assumption of the listener calculation. This is a model-based extension to counterfactual referents within the observed task context.

Let $S_{it}^{(d)}(u \mid o, X)$ be the resulting complete-response probability at posterior draw d. We evaluate the speaker's size-first probability conditional on the size–colour adjective set as

$$\frac{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X)}{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X) + S_{it}^{(d)}(\mathrm{CS} \mid o^*, X)}. \tag{25}$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval. Replaying the original target reproduces its stored likelihood within 10^{-8} for every model.

Figure 20 reports the resulting speaker and listener predictions. In the participant-weight plan-guided models, the mean size-first target-probability advantage was .0062 under fixed semantics, 95% posterior interval [.0047, .0077], and .0058 under updating, [.0045, .0072]. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were .967 and .966. Both fitted regimes therefore generate a size-first advantage under this counterfactual listener calculation. In the response-score decomposition, the predictive difference between these models was concentrated in adjective content and length. The two calculations describe how

the models differ in predicting the choice of a description while both assigning a listener advantage to size-first order.

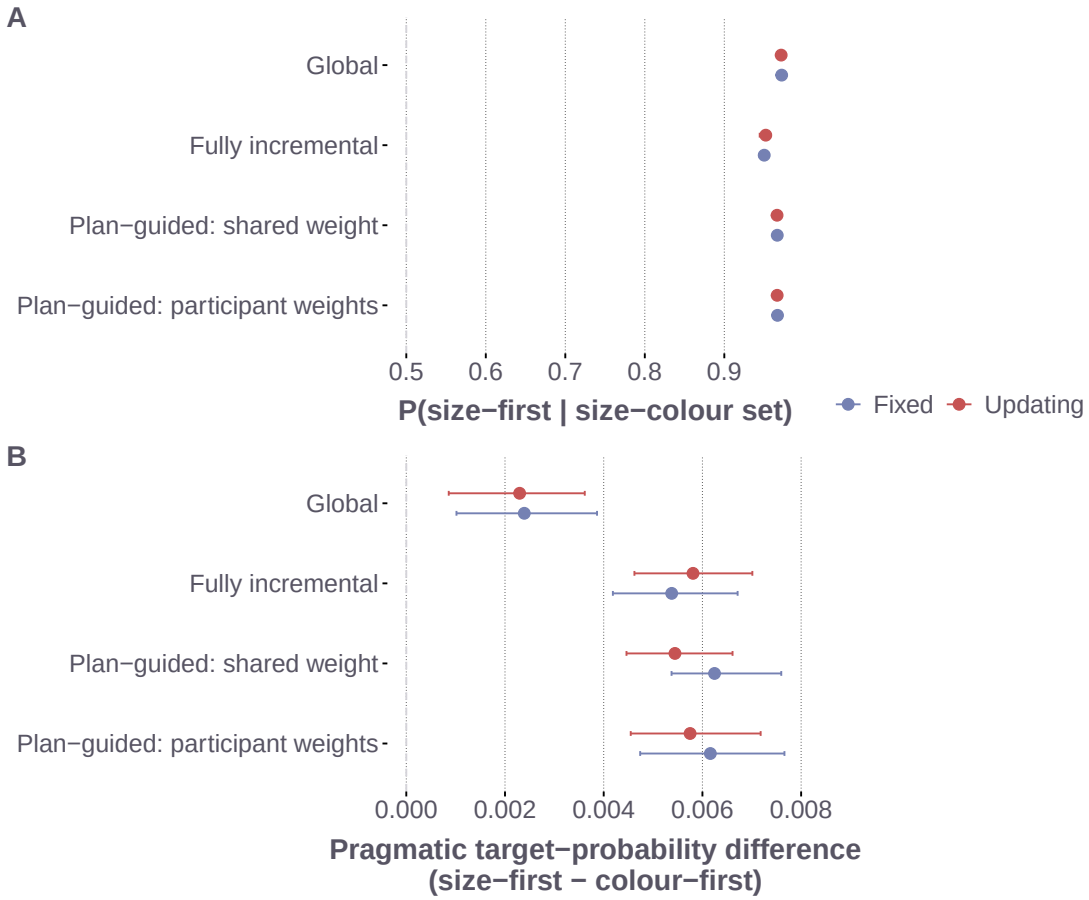


Figure 20: Supplementary predictions from the fitted production models on the size–colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size–colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 posterior parameter draws per model.